

PROVISIONAL PATENT APPLICATION

Winding-Register Memory Cells, Frequency-Addressed Winding-Register Memory Devices, and Sink-Gated Resonant Power-Switching and Receiver-Keyed Signaling Devices Built on a Resonantly Driven Ordered Medium, and Methods of Writing, Reading, Refreshing, Qualifying, and Operating Them

Inventor: Christopher F. Milam

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This specification is complete and self-contained. Applicant's companion applications are nonessential background only; no material essential to written description, enablement, or the claims is incorporated from them. Every example under PROPHETIC EXAMPLES is prospective; no experiment, retained state, retention time, level count, readout resolution, or signaling figure is represented as achieved. Where a retention channel longer than the medium's population relaxation is discussed it is identified expressly as the applicant's hypothesis, and the products are disclosed so as to be buildable and operable whether or not that hypothesis is confirmed at qualification.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is one of a coordinated series directed to applications of controlled angular-momentum coherence in condensed media (companion drafts: support-force control apparatus; transportation and lift systems; measurement instruments; energy storage and infrastructure; media and array architectures; time-rate metrology; and completion-scheduled computation). Each companion states its own relationship to this disclosure and remains self-contained as to claim-essential matter.

FIELD

Information storage, memory devices, and resonant signaling. More particularly: (i) a **winding-register memory cell** whose stored quantity is a committed count of drive- synchronous whole turns deposited by counted resonant pulses into the aligned internal angular-momentum population of an insulating single-crystal medium, held in a tuned coupling structure under an alignment field, and read back non-destructively through a defined winding observable (spectroscopic, force-channel, or channel-partition); (ii) a **frequency-addressed winding-register memory device** in which a plurality of such cells on one drive feed are addressed by their bias-set resonance frequencies; (iii) a **sink-gated resonant power-switching device** and (iv) a **receiver-keyed resonant signaling device**, in which delivery on an established resonant linkage is gated and keyed by the presentation, withdrawal, or detuning of the receiving resonator; (v) the **deployed operating methods** (write, read, verify, refresh, erase; power

gating; receiver-keyed symbol transfer) by which such devices are run from a stored retention and readout map; and (vi) the **factory qualification** by which the map—including the measured hold time of a cell class and the identification or non-identification of a long-retention channel—is established.

BACKGROUND

1. **Spintronics** (spin-transport electronics; GMR read heads, MRAM, STT-MRAM/SOT-MRAM) stores information in the *orientation* of spin populations and reads it electrically through charge transport (magnetoresistance). Its state variable is directional — binary or few-level — and its architectures route information through charge carriers, inheriting scattering loss and treating precession coherence chiefly as a nuisance to switching. The field's own write physics concedes the mechanical nature of the stored quantity: spin-transfer-torque switching obeys $dL/dt = \eta(\hbar/2)(I/e)$ — each carrier delivers up to $\hbar/2$ of angular momentum — and on public device values (a $40\text{ nm} \times 1.3\text{ nm}$ CoFeB free layer at $M_s = 10^6\text{ A/m}$ holds $\approx 1.8 \times 10^5$ units of $\hbar/2$; $50\text{ }\mu\text{A}$ at $\eta \approx 0.6$ delivers 1.9×10^{14} units per second) the conservation arithmetic returns $t \approx N \cdot e / (\eta I) \approx 0.9\text{ ns}$ to deliver the layer's count once, and $\approx 1.9\text{ ns}$ for a full reversal delivering $2N$ units ($+N \rightarrow -N$) — the datasheet class (the actual trajectory is governed by the damped precession equation; the count arithmetic fixes the order). The industry already writes by counted angular-momentum delivery while narrating the stored state as a two-valued label.
2. **Magnonics** carries information in propagating spin waves in magnetic insulators — notably yttrium iron garnet, adopted by that field for the same narrow-linewidth coherence exploited here (Gilbert damping $\alpha \approx 3 \times 10^{-5}$ in YIG, the lowest known in any magnetic material; centimetre-class propagation at gigahertz rates) — and reads it inductively or magneto-optically. Its state variable is the presence/phase of a traveling excitation; storage is transient.
3. Neither field possesses, nor suggests, the state variable and readouts disclosed here: (a) **winding depth** — a committed count of drive-synchronous whole turns deposited into an aligned population by counted resonant pulses and read back as a multi-level register value through a defined observable (not an orientation, not a traveling wave); (b) **force-channel readout** — reading the stored register value as a differential change in the support force of the storage body under a drive-toggle protocol and a calibrated read gradient, a readout in which the cell emits no readout signal and which is performed through electromagnetically opaque containment; (c) **channel-partition readout** — the operational response class of the driven population under alignment-sense reversal, usable as an orthogonal stored symbol dimension where the qualified cell class exhibits at least two drive-selectable classes.
4. The applicant's framework identifies why these were left on the table: the extant fields treat spin as a two-valued label with an attached magnetic moment, rather than as the mechanical angular-momentum organization it demonstrably is (Einstein–de Haas; Barnett). Orientation is one bit of a far richer state space: a coherent population also carries how far it has been wound (quantized in whole turns), which internal channel holds the deposit, and a mechanical constraint-force signature. The disclosure occupies that space.

5. The taxonomy makes the vacancy precise. Every deployed memory stores one of two quantities. An **orientation** — which way an aligned population points (hard-disk grains, MRAM free layers): a *state*, persisting without upkeep on an Arrhenius barrier. Or a **bias** — a counted census held off balance (DRAM at $\approx 1.9 \times 10^5$ elementary charges per 30 fF cell, refreshed every 64 ms per the JEDEC standard because the census leaks; flash, the same census walled behind a ≈ 3.2 eV oxide barrier). Orientation stores persist because a completed organization is a state, not a process; bias stores decay because a displaced census is a process, not a state. The register disclosed here — a committed **winding depth** of a driven aligned population — is operationally a *counted census* like the second family, but one written and read without charge transport, without a junction, and without a transistor in the storage medium, with a level count set by count resolution rather than by a two-valued barrier. Whether any component of that census is retained on a channel longer than the population's own relaxation (a state-like component) is the applicant's hypothesis of this disclosure; it is stated as such, and a protocol that would identify it or bound it is disclosed. The baseline product is refresh-scheduled from the measured hold time and does not depend on the hypothesis.

Applicant's published exhibit corpus (nonessential background). Applicant maintains a public, commit-dated exhibit series presenting the underlying research program in didactic depth (the Normal Realism / POAMS exhibits, <https://cfmilam.github.io/nr-poams-exhibits/>). Those pages are applicant's own prior publications and are cited as nonessential background teaching only: no material essential to enablement, written description, or the claims is incorporated from them, and this specification is complete and self-contained.

SUMMARY

The disclosure is organized in three tiers (FIG. 13). The products and their deployed operating methods lead; the factory qualification establishes the retention and readout map each product runs from; the validity, recalibration, and safe-off provisions govern what a deployed unit may do.

Tier 1 — products and deployed operating methods (leading). (a) A **winding-register memory cell** (700; FIG. 1) comprises: a body of ordered medium (702) of sufficient phase coherence in a holder (703); an alignment-field source (704) establishing measured saturation alignment and a substantially uniform field over the body during write and hold; a coupling structure (706) and a locked pulsed drive chain (708) at the population's located precession resonance; a spectroscopic probe port (710); a controller (712) holding a **winding register** (714) and a **calibration memory** (716) storing the cell's **retention and readout map**; a **refresh scheduler** (718); a safe-off state (720); and, in species, a force-channel readout (730: force sensor 732, calibrated read-gradient source 734, electromagnetically opaque containment 736 with drive feedthrough 738), a channel-partition readout (740), and a cryogenic enclosure (750). WRITE: counted resonant pulses advance the collective precession phase by a controller-defined number of whole turns per level, the committed count being the stored **winding depth**. HOLD: bias-maintained retention for the qualified hold time of the cell class, with refresh (read-then-rewrite at read power) scheduled from

the map. READ: the **winding observable**—non-destructive absorption spectroscopy of line shift and saturation depth; or differential support force under a drive-toggle protocol with a calibrated read gradient; or channel-partition response class—mapped to the level index by the readout map. VERIFY: read-back compared with the committed count within the level tolerance. ERASE: commanded counter-sense drive (coherent de-excitation, assuming phase coherence of the retained component) or a disclosed relaxation pulse (bias-field detuning through a fast-relaxing crossing; or a temperature pulse). (b) A **frequency-addressed winding-register memory device** (800; FIGS. 7–8): N cells on a common drive feed, each biased by a stepped, piecewise-uniform bias (permanent-magnet shim species—no per-cell conductors; or trim-coil species) to a distinct resonance address, with a disclosed **guard-band rule**, pulse-bandwidth rule, in-situ address calibration, guard cells, shielding partitions, and weakly coupled feed taps as the crosstalk architecture; block operations by comb waveforms. (c) A **sink-gated resonant power-switching device** (900; FIG. 11): a source readiness apparatus, a linkage gate, and delivery to a load only when a matched resonant sink completes the linkage. In the POAMS-native embodiment no transferable content, delivery, or separately propagated enable signal exists before completion; optional pilot resonator instrumentation is an implementation/control species, not a prerequisite of the endpoint-completion architecture. (d) A **receiver-keyed resonant signaling device** (950; FIG. 12): the sink encodes symbols by modulating its own readiness; the source reads the completed or uncompleted exchange from its local delivery ledger. No signal is emitted absent a completed linkage in the POAMS-native embodiment. Optional antenna or pilot measurements are artifact controls only.

Tier 2 — factory qualification. For each cell class the qualification apparatus (FIG. 10) executes the write-and-observe protocol at each level over log-spaced observation windows, the **multi-timescale retention diagnostic**, and the readout calibration, and stores the retention and readout map: the level-to-observable relation with uncertainty; the read-disturb threshold; the measured hold time constant τ_{hold} of the retained component; whether a **long-retention channel** passed the identification gates R1–R6 (and its τ_{long} as a function of temperature and bias) or only an upper bound; the refresh interval computed by the refresh rule; validity envelope; provenance; safe-off parameters. The cell class is graded **register-class** (hold on the population relaxation; refresh scheduled accordingly) or **memory-class** (hold on an identified long-retention channel). Both are products of this disclosure.

Tier 3 — validity, recalibration, safe-off. Validity states (valid, degraded, out-of-envelope, invalid, safe-off); interleaved zero and reference reads; drift-triggered envelope shrinkage or recalibration; the rule that a deployed unit never schedules a refresh interval, level count, or readout its map does not support.

Layered scope (FIG. 13). Every quantity is disclosed at three layers: (L0) the raw synchronous observable—the change in the probe, force, or telemetry channel synchronized with a commanded write, read, or readiness state, always reportable; (L1) the mapped register value—the level index with uncertainty, reportable inside the validity envelope; (L2) operation—write/read/verify/refresh/erase, addressing, gating, and symbol transfer from the map. No layer presumes a retention time, level count, readout resolution, sign, or scaling beyond the validated points; the map admits a register-class grade.

Reference hardware. A polished single-crystal yttrium-iron-garnet sphere (array grade 0.2–2 mm, reference 1 mm; bench and force-readout grade 2–15 mm, reference 10 mm) at a mapped transverse microwave magnetic-field antinode of a rectangular TE₁₀₂ coupling structure at 5.600 GHz (empty-cavity seed approximately $47.55 \times 22.15 \times 64.77$ mm) or at a weakly coupled tap of a shared transmission-line feed; a permanent-magnet yoke or C-frame electromagnet at 1500–2500 Oe; a locked pulsed drive chain (4–8 GHz, pulse width 0.1–10 μ s, controller-enforced duty bound); a swept or dual-tone probe; and a controller with calibration memory.

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Winding-register memory cell (700): ordered medium in holder, alignment-field source, coupling structure and locked pulsed drive chain, spectroscopic probe port, controller with winding register, calibration memory holding the retention and readout map, refresh scheduler, safe-off; force-channel readout species with read-gradient source and containment feedthrough; channel-partition readout species; cryogenic enclosure.
- FIG. 2 — State variable and readout definition: per-pulse increment $\rightarrow$ whole turn $\rightarrow$ level $\rightarrow$ committed winding register; the three winding observables (spectroscopic line shift and saturation depth; differential support force under drive toggle with read gradient; channel-partition response class) and the readout map that returns the level index.
- FIG. 3 — Write timing: counted pulse train; committed winding depth advancing one level per $n_{\text{turn}} \cdot m$ pulses; level ladder 0 ... L-1.
- FIG. 4 — Spectroscopic readout: probe absorption line at levels 0, m, 2m; line-center shift and saturation depth versus stored level; read power below the read-disturb threshold.
- FIG. 5 — Force-channel readout: read-power drive toggle windows with the calibrated read gradient applied; $\Delta F = \text{mean}(\text{ON}) - \text{mean}(\text{OFF})$; monotone map from stored level to ΔF with the seven artifact controls.
- FIG. 6 — Channel-partition symbol plane: stored symbol = (winding depth, response class); classes assigned by the classification map; λ reported only with endpoint standards.
- FIG. 7 — Frequency-addressed winding-register memory device (800): common feed with weakly coupled taps; cells at addresses $f_1 \dots f_N$ set by stepped bias; address map with guard bands.
- FIG. 8 — Crosstalk architecture: guard-band rule ($\text{pitch} \geq 3\gamma\Delta H + \text{pulse bandwidth}$), piecewise-uniform stepped bias, guard cells, shielding partitions, in-situ address calibration, dipolar-shift budget.
- FIG. 9 — Retention (prospective): retained winding depth versus log time for register-class and memory-class grades; half-level misread threshold; refresh interval $t_{\text{ref}} \leq \tau_{\text{hold}}/(2L) \cdot s$.
- FIG. 10 — Refresh-scheduled register and retention-identification protocol: write $\rightarrow$ observe on log-spaced windows $\rightarrow$ fit cavity ring-down, population ring-down, thermal relaxation, residual $\rightarrow$ gates R1–R6 $\rightarrow$ map entry (τ_{long} or bound) $\rightarrow$ refresh rule $\rightarrow$ grade.

- FIG. 11 — Sink-gated resonant power-switching device (900): source readiness apparatus, linkage gate, resonant sink; delivery only upon completed linkage; optional readiness-hardware loss and instrument registration time reported separately from the exchange.
- FIG. 12 — Receiver-keyed signaling device (950): sink readiness symbol stream; linkage state; source delivery telemetry reading the symbols; standby radiated power below the stated limit.
- FIG. 13 — Three-tier architecture and layered scope: Tier 1 products and deployed operating method (write/read/verify/refresh/erase state machine); Tier 2 factory qualification producing the retention and readout map and the cell-class grade; Tier 3 validity states, recalibration, safe-off; layers $L0 \rightarrow L1 \rightarrow L2$.

DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

Portfolio operating terminology. "POAMS-native" and "direct-mode" identify the prophetic operating tier that applies the disclosed POAMS bookkeeping directly rather than requiring a preexisting nonzero empirical response map. "Commanded" and "selectable" identify controller- addressable input states and do not promise a nonzero output. "Independently validated" means reproduced under the disclosed qualification protocol against a separate reference; "empirically established" means recorded with uncertainty and provenance in a stored response map. Extant-framework labels appearing in quotation marks or introduced by "extant label:" identify measured phenomena by their customary names; their use commits neither the specification nor the claims to any entity ontology historically associated with those names. Every measured claim quantity is fixed by a recited instrument reading. Hypothesis parameters are identified expressly and are not presumed to have independently measured values.

1. **Ordered medium / aligned internal angular-momentum population** — a body whose constituent angular momenta can be aligned by a static field, measured in aggregate as magnetization (A/m; standard magnetometry); extant label "electron spin." Reference medium single-crystal yttrium iron garnet ($Y_3Fe_5O_{12}$, YIG).
2. **Alignment field; substantially uniform** — the static bias field (oersted or tesla) applied to establish measured saturation alignment; "substantially uniform" means that the field variation across the body is less than the address pitch expressed in field (Definition 22) divided by three, and that the force-axis gradient is within the gradient budget of the readout in use. The **read gradient** (Definition 12) is a deliberate, calibrated departure applied only during force-channel readout.
3. **Precession resonance** — the absorption resonance of the aligned population under the alignment field (extant label: ferromagnetic resonance, FMR; $f = \gamma H_{eff}$, $\gamma \approx 2.8$ MHz/Oe electron-class), located operationally as the absorption extremum to which the drive servo locks.
4. **Population relaxation time T_1 ; population coherence time T_2^* ; cavity energy-decay time** — T_1 is the decay time of the excited precession amplitude (cone angle / excited population) after drive termination, measured by pulsed response; for Gilbert damping α at angular frequency ω , T_1

$\approx 1/(2\alpha\omega)$ (reference medium: $\alpha \approx 3 \times 10^{-5}$ at 5.6 GHz gives $\approx 0.47 \mu\text{s}$). T_2^* is the coherence decay time determined from the resonance linewidth, $T_2^* \approx 1/(\gamma\Delta H)$ ($0.11\text{--}0.28 \mu\text{s}$ for $\Delta H = 0.5\text{--}0.2 \text{ Oe}$). The cavity energy-decay time $\tau_c = Q_L/\omega$ is measured from loaded Q and ring-down and is not identified with either.

5. **Sufficient phase coherence** — $T_2^* \cdot \nu \geq 1$ at drive turn rate ν ; preferred embodiments exceed this floor by two to three orders of magnitude (single-crystal YIG at gigahertz drive, to be measured for the specimen and mode).
6. **Winding increment ($\delta\phi$)** — the drive-synchronous advance of the collective precession phase deposited by one resonant pulse, $\delta\phi = \gamma h_1 \tau_p$ (rotating-frame nutation for pulse amplitude h_1 and width τ_p) at the cell's operating point, as established by the readout calibration; a **whole turn** is 2π of committed increment, $n_{\text{turn}} = \lceil 2\pi/\delta\phi \rceil$ pulses.
7. **Level; level count L; pulses per level** — a level is a controller-defined integer multiple m of whole turns; the pulses per level $n_L = m \cdot n_{\text{turn}}$ are fixed in the map for the cell class; L is the number of distinguishable levels the map supports at the stated confidence.
8. **Winding depth (W); winding register** — W is the committed count, since the last erase, of whole turns deposited into the cell by counted pulses, expressed as a level index $0 \dots L-1$; the *winding register* (714) is the controller record of the committed count. The **retained winding depth** is the level index returned by a winding observable through the readout map at read time. W is a material state only to the extent the readout map returns it; that extent is measured at qualification, not presumed.
9. **Winding observable** — an instrument reading that the readout map relates to the level index: (a) the *spectroscopic observable*—the probe absorption line's center shift Δf and saturation depth relative to the level-0 reference at read power below the read-disturb threshold; (b) the *force-channel observable*—the differential support force $\Delta F = \text{mean}(F_{\text{ON}}) - \text{mean}(F_{\text{OFF}})$ of the body under a read-power drive toggle with the calibrated read gradient applied; (c) the *channel-partition observable*—the sense-odd and sense-even components of ΔF under alignment-sense reversal (Definition 15).
10. **Readout map** — the stored relation, with uncertainty, between each winding observable and the level index for the cell class, including the level-0 reference, full-scale observable, monotonicity record, read-disturb threshold, and read-power and averaging settings.
11. **Read-disturb threshold** — the probe or toggle power at which N_r consecutive reads (N_r stored in the map, reference 100) change the returned level by less than one half of a level; reads are performed below it.
12. **Read gradient** — a calibrated force-axis field gradient $\partial B/\partial z$ applied by the read-gradient source (734) only during force-channel readout, of stored magnitude and reversible sign, converting a retained change ΔM_z of the body's moment into a force $F = m_{\text{body}} \cdot (\Delta M_z/M_s) \cdot \partial B/\partial z$; absent during write and hold.
13. **Hold time constant (τ_{hold}); long-retention channel (τ_{long}) — hypothesis** — τ_{hold} is the measured time constant of the level-dependent component of the winding observable after write,

whatever its carrier. The applicant hypothesizes that a component of the committed winding is retained on a *long-retention channel* with $\tau_{\text{long}} \gg T_1$, identified only by the protocol of Definition 14; candidate carriers named as hypotheses are (i) a metastable collective wound state of the aligned electron population, (ii) transfer to the nuclear-spin population of the medium through hyperfine coupling, and (iii) mechanical angular momentum of the body on a compliant mount. Absent identification the map records an upper bound on τ_{long} and $\tau_{\text{hold}} = T_1$ of the population.

14. **Retention-identification gates R1–R6** — the acceptance gates the multi-timescale diagnostic must pass before τ_{long} is entered in the map: R1 *time separation*—a level-dependent residual persists beyond $100 \cdot T_1$ and $10 \cdot \tau_{\text{th}}$ (thermal relaxation of the body); R2 *resonance contrast*—absent under an absorbed-energy-matched detuned write; R3 *material contrast*—absent for a geometry-, mass-, dielectric-matched non-alignable body; R4 *count monotonicity*—residual amplitude increases monotonically with committed count over at least three levels with consistent sign; R5 *erase contrast*—removed by a commanded erase and not by an equal elapsed time; R6 *replication*—reproduced on a second cell, apparatus, or laboratory within stated uncertainty.
15. **Sense-odd / sense-even components; response class; endpoint standards; channel partition (λ)** — $\Delta F_{\text{odd}} = (\Delta F_{+} - \Delta F_{-})/2$ and $\Delta F_{\text{even}} = (\Delta F_{+} + \Delta F_{-})/2$ from the force-channel observable at the two alignment senses in separate settled blocks. A *response class* is an operational label assigned by a stored classification map from the signed, normalized (ΔF_{odd} , ΔF_{even}) vector and its uncertainty (for example: sense-odd positive; sense-odd negative; sense-even dominated; null; unclassified). *Endpoint standards* are independently established response-class standards for an intrinsic-alignment-dominated ("twist") route and an orbital-winding-dominated ("writhe") route; the *channel partition* λ is a hypothesis parameter reportable as an estimate $\hat{\lambda}$ only when endpoint standards are present. Response classes are defined by measurement, not by physical account; no sign is presumed.
16. **Refresh; refresh rule** — refresh is a read at read power followed by rewrite of the returned level. The refresh interval is $t_{\text{ref}} \leq s \cdot \tau_{\text{hold}}/(2L)$, with $s \leq 1$ a stored safety factor (reference 0.5), because a linearly decaying observable is misread by half a level at $t \approx \tau_{\text{hold}}/(2L)$.
17. **Register-class; memory-class; qualified cell class** — a *qualified cell class* is a cell design (medium property box, geometry, coupling structure, bias, drive chain, probe) represented by qualified units whose calibration memory holds a retention and readout map with uncertainty, provenance, validity envelope, refresh rule, and safe-off parameters. It is *register-class* if τ_{hold} is the population relaxation (no long-retention channel identified) and *memory-class* if a long-retention channel passed R1–R6; the grade is a measured output. Neither grade presumes level count or retention beyond the map.
18. **Retention and readout map (stored)** — the data structure held in the calibration memory (716): readout map (Definition 10); τ_{hold} and, where identified, $\tau_{\text{long}}(T, H_0, \text{level})$ with uncertainty or the recorded bound; grade; refresh rule parameters; level count L ; pulses per level; write pulse

parameters; validity envelope; provenance record; safe-off parameters; zero-read and reference-read history.

19. **Provenance record; validity envelope; runtime witnesses** — as for any qualified module: unit or class identity, qualification apparatus and references, date, raw-data identifier, fit method, gates passed; the region of drive, bias, temperature, orientation, gradient, and vibration over which the map was established, with runtime limits; and the reduced sensor set (body and mount temperature, bias magnitude, force-axis gradient witness, incident/reflected/absorbed drive power, lock state, vibration) monitored at runtime.
20. **Validity states; safe-off** — *valid, degraded, out-of-envelope, invalid, safe-off*; safe-off is drive off, bias to stored safe magnitude (or unchanged in a fixed-bias unit), read gradient off, winding register and last valid read preserved, recalibration flag set.
21. **Medium property box** — the stated ranges of measured medium properties for a cell class, with measurement conditions (Species table in the description).
22. **Address; address pitch; guard band** — the address of a cell in a frequency-addressed device is its bias-set resonance frequency f_i ; the pitch is $f_{i+1} - f_i$ (in field, pitch/ γ); the guard band is the pitch less the occupied linewidth.
23. **Linkage; sink readiness; completed linkage** — the condition in which a resonant load ("sink") is presented and matched to a source resonator such that delivery proceeds (extant labels: matched load, critical coupling); readiness is presented, withdrawn, or detuned by the sink.
24. **Hand-off rate** — the delivered power on the linkage expressed as a count rate of action quanta, $\dot{N} = P_{\text{del}}/h\nu$ at linkage turn rate ν ; modulation of hand-off rate is modulation of delivered power by the sink's coupling state.
25. **Readiness; completed linkage; standby** — readiness is the source-side and sink-side configuration capable of co-signing an exchange; a completed linkage is the two-ended event in which both readiness conditions close and a hand-off is booked. In the native endpoint-completion embodiment no transferable content or separately propagated enable signal occupies the interval before completion, and standby delivery is zero. A pilot resonator, stored pilot energy, antenna monitor, or listening-band limit may be used as an optional extant-compatible control species but does not define the linkage.

THE WINDING-REGISTER MEMORY CELL (700) — PRODUCT ARCHITECTURE (FIG. 1)

Ordered medium (702) and holder (703). The reference medium is a polished single-crystal YIG sphere carried in a low-loss, nonmagnetic holder (quartz, sapphire, PTFE, alumina). Two reference species and two shape alternatives are disclosed; each cell class is qualified on its own measurements.

Medium property box (Species table). Measurement conditions: FMR linewidth ΔH is the full width at half absorption at the operating frequency (reference 5.600 GHz), 295 K, probe power below the read-disturb threshold; resistivity by two-terminal DC measurement at 295 K; saturation induction by

magnetometry at 295 K.

Species	Geometry	ΔH	ρ	$4\pi M_s$	Use
S (array grade)	sphere 0.2–2 mm (ref. 1 mm; mass ≈ 2.7 mg)	≤ 0.5 Oe (≤ 1.4 MHz)	$\geq 10^{10}$ $\Omega \cdot \text{cm}$	$\approx 1780 \text{ G} \pm 5\%$	frequency-addressed devices; array-consistent with a 5 MHz pitch
L (bench / force-readout grade)	sphere 2–15 mm (ref. 10 mm; mass ≈ 2.71 g; moment $\approx 0.074 \text{ A} \cdot \text{m}^2$)	≤ 10 Oe	$\geq 10^{10}$ $\Omega \cdot \text{cm}$	$\approx 1780 \text{ G} \pm 5\%$	single cells; force-channel and channel-partition readout
D (shape-pulling grade)	disk or rod, aspect $\geq 5:1$, volume-equivalent to S or L	≤ 2 Oe	$\geq 10^{10}$ $\Omega \cdot \text{cm}$	$\approx 1780 \text{ G} \pm 5\%$	high-resolution spectroscopic readout via shape-field pulling
C (cryogenic grade)	any of S/L/D	measured at 77 K and, optionally, 4 K	—	measured at temperature	cryogenic enclosure (750)

Substituted iron garnets (Ga-, Al-, La-, Bi-) and other insulating ordered media of sufficient phase coherence are candidate media qualified on their own measurements; films, polycrystals, and composites are separately characterized, never presumed equivalent.

Alignment-field source (704). A permanent-magnet yoke with shim elements (the conductor-free species), a permanent-magnet yoke with a trim winding, or a C-frame electromagnet (reference 76 mm-pole class, approximately 0.196 T at 20 A and an 80 mm gap), producing 1500–2500 Oe at the body with the uniformity of Definition 2. During write and hold the field is substantially uniform; the sign of any response is taken from the map, never from field polarity. Hold energetics: a permanent-magnet species holds alignment with zero standing power; an electromagnet species has a standing coil dissipation (reference ≈ 0.20 kW for the C-frame class) that is a stated cost of that species.

Coupling structure (706) and drive chain (708). A TE-mode rectangular cavity (reference TE₁₀₂ at 5.600 GHz, empty-cavity seed approximately $47.55 \times 22.15 \times 64.77$ mm; in the ideal empty-cavity field map the microwave magnetic field remains $\geq 88.5\%$ of its central value across a 10 mm body), loop-gap resonator, re-entrant resonator, stripline or coplanar resonator, or a weakly coupled tap of a shared transmission line (array species; coupling rate g much less than the line's loss rate so that no cavity–population hybrid doublet forms) places the body at a mapped transverse microwave magnetic-field antinode with the drive's magnetic axis orthogonal to the alignment field. The drive chain comprises a voltage-controlled source spanning the resonance (reference 4–8 GHz), a pulse and burst modulator with controller-enforced duty bound (pulse width 0.1–10 μs , repetition 50 Hz–1 MHz in nonoverlapping combinations; reference write pulse 2 μs , giving a 0.5 MHz spectral width), a power amplifier sized to the map (reference ≤ 10 mW peak at the cell for species S; 1–5 W for species L at qualification), a directional coupler, and incident/reflected/transmitted power detectors. The controller servo-locks to the located precession resonance and records mode, linewidth, loaded Q , and ring-up/ring-down times; $Q \times$ source power is not treated as a field multiplier.

Spectroscopic probe port (710). A swept-frequency or dual-tone probe at read power below the read-disturb threshold, reading reflected or transmitted power through the coupling structure; line-center estimation by fit to the recorded absorption dip; saturation-depth estimation by comparison of the dip depth with the level-0 reference at equal probe power.

Controller (712), winding register (714), calibration memory (716), refresh scheduler (718), safe-off (720). The controller executes the deployed operating method (below), maintains the winding register, applies the readout map, propagates uncertainty, enforces the validity envelope and refresh rule from the calibration memory, and executes safe-off. It refuses a map whose provenance does not match the unit or class identifier.

Force-channel readout species (730). A force sensor (732: flexure with capacitive or optical readout, load cell, or balance; reference threshold 0.01 mg-force $\approx$ 98 nN) supports the holder through a defined body load path separate from the reaction path that carries the coupling structure, alignment source, feeds, and containment. A read-gradient source (734: a pair of trim coils or a movable permanent-magnet pair on the reaction frame) applies the calibrated read gradient of Definition 12 during readout only. Where the cell is sealed in electromagnetically opaque containment (736: copper or aluminium enclosure), the drive enters through a filtered feedthrough (738: coaxial feedthrough with below-cutoff waveguide filter, or an optical fibre to an internal photodiode-driven oscillator) or is generated by an internal source with internal power; the *readout* passes through the wall by weighing alone.

Channel-partition readout species (740). The force-channel species with an alignment-field source of commanded sense, a controller executing alignment-sense reversal in separate thermally settled blocks, and a classification map (Definition 15) in the calibration memory.

Cryogenic enclosure (750). A cryostat on the reaction frame; the body load path (where a force sensor is used) passes through a thermally anchored, low-conductance stem; boil-off, pressure fluctuation, thermal-strap forces, and settling time are witnessed and gated. Cooling is not presumed to narrow the line or lengthen τ_{hold} ; both are measured at each temperature and entered in the map.

STATE VARIABLE AND READOUT (FIG. 2)

A resonant pulse of amplitude h_1 and width τ_p at the located precession resonance advances the collective precession phase of the aligned population by the winding increment $\delta\phi = \gamma h_1 \tau_p$ in the rotating frame. In steady-state resonance the orientation is periodic modulo 2π and cycle count is not by itself a retained material state; the committed count is retained in the winding register (714), and its material counterpart is whatever the winding observable returns. The excited precession amplitude (cone angle θ ; equivalently the reduction $\Delta M_z = M_s(1 - \cos \theta)$ of the aligned moment, or an excited-population number) is the level-dependent quantity a counted pulse train deposits; it relaxes on T_1 unless a long-retention channel is identified (Definition 13). The disclosure therefore defines the stored level by the committed count and reads it through a calibrated observable; it does not presume that the observable persists beyond the measured τ_{hold} .

Spectroscopic observable and mechanism species. The precession frequency depends on the aligned moment through anisotropy and shape fields. For a sphere the shape term cancels and cubic anisotropy (reference $H_a \approx 40\text{--}80$ Oe in YIG) pulls the line by $\Delta f \approx \gamma H_a (\Delta M_z / M_s) \approx 2.8 \text{ MHz/Oe} \times 60 \text{ Oe} \times 0.01 \approx 1.7 \text{ MHz}$ per 1 % of M_s ; for species D the shape term $\gamma \cdot 4\pi M_s \cdot \Delta N$ (gigahertz-class $\times \Delta M_z / M_s$) is one to two orders larger, which is the high-resolution embodiment. Saturation depth—reduced probe absorption while the population is already excited—is a second observable, present only while the excited component persists. Level resolution: the line-center precision $\sigma_f \approx \Delta f_{\text{line}} / (\text{SNR} \cdot \sqrt{M})$ for M averaged sweeps at signal-to-noise SNR must satisfy $\sigma_f \leq \Delta f_{\text{FS}} / (5L)$ for adjacent levels at $\geq 5\sigma$, where Δf_{FS} is the full-scale shift of the map; the observable dynamic range required is $20 \log_{10}(5L)$ dB (62 dB for $L = 256$; 86 dB for $L = 4096$).

Force-channel observable. Angular-momentum delivery produces torque, not net vertical force, on a rigidly held body; the disclosed transduction is the gradient force on a retained change of the body's moment, $F = m_{\text{body}} \cdot (\Delta M_z / M_s) \cdot \partial B / \partial z$ under the calibrated read gradient (species L: $m_{\text{body}} \approx 0.074 \text{ A} \cdot \text{m}^2$; $\Delta M_z / M_s = 1 \%$ at 0.01 T/m gives $\approx 7.4 \text{ } \mu\text{N} \approx 0.76 \text{ mg-force}$, 76 times the 98 nN reference floor). The applicant's further hypothesis that the retained winding itself alters the body's support force is not relied on; the map records whatever level-dependent ΔF passes the controls. **Seven controls** are part of the readout method: (1) off-resonance toggle at equal power; (2) matched non-magnetic dummy body; (3) alignment- sense reversal—a gradient-force signal flips with $\partial B / \partial z$, artifacts do not; (4) read-gradient sweep—signal $\propto \partial B / \partial z$ if magnetic; (5) interleaved, blinded level-0 versus level- k reads; (6) toggle-frequency scan across the thermal time constant; (7) balance-EMI check with the body removed. Thermal artifact scale (species L, 10 mW absorbed for 1 s into $\approx 1.6 \text{ J/K}$ gives $\approx 6 \text{ mK}$; $dM/dT \approx -0.05$ to -0.1% /K gives $\Delta M / M \approx 3\text{--}6 \times 10^{-6}$; at 0.01 T/m that is $2\text{--}5 \text{ nN} \approx 0.2\text{--}0.5 \text{ } \mu\text{g-force}$) is recorded and bounded below one third of the level spacing in force.

Channel-partition observable. Definition 15; the classification map is established at qualification (FIG. 6). The second symbol dimension exists for a cell class only if the map shows at least two drive-selectable response classes (selected by pulse width, bias point, or drive polarization) that are distinguishable at the stated confidence; otherwise the cell stores winding depth alone.

DEPLOYED OPERATING METHOD (FIG. 13, Tier 1)

Performed by the controller (712) of a cell whose calibration memory holds a retention and readout map:

(a) *Power-on and self-test.* Load the map; verify provenance; establish alignment; locate and lock the resonance; perform a level-0 reference read; enter *valid* only if within the map's zero tolerance. (b) *Write.* Receive a target level k ; apply $k \cdot n_L$ counted pulses at the map's pulse parameters (or the difference from the current register value for incremental write); update the winding register. (c) *Verify.* Read below the read-disturb threshold; compare the returned level with the register within one half level; on mismatch rewrite once, then flag *degraded*. (d) *Hold.* Maintain alignment; read gradient off; monitor runtime witnesses. (e) *Refresh.* At the refresh interval of Definition 16, read and rewrite the returned level (read-then-rewrite); log the drift statistics. (f) *Read.* Return the level index with uncertainty from the readout

map; where the force- channel species is used, apply the read gradient during the toggle windows only and remove it after the read. (g) *Erase*. Commanded counter-sense drive of the committed count (coherent de-excitation, valid when the retained component is phase-coherent); or a relaxation pulse—a bias-field excursion detuning the population through a fast-relaxing crossing, or a controlled temperature pulse—after which a level-0 reference read confirms erasure. Irreversible erasure of one bit dissipates no less than $kT \ln 2$ (300 K: 2.87×10^{-21} J, a 4.33 THz quantum, ≈ 774 quanta at 5.6 GHz); no step claims sub-Landauer operation. (h) *Validity and safe-off*. On a runtime-limit exceedance, loss of lock, drift beyond the map tolerance, or fault, transition to *safe-off* (Definition 20) and set the recalibration flag; the winding register and last valid read are preserved. The operator of a deployed unit does not perform the qualification protocol; R1–R6 and the readout calibration are executed once per cell class and again only on service recalibration.

RETENTION: HYPOTHESIS, IDENTIFICATION PROTOCOL, AND REFRESH-SCHEDULED BASELINE (FIGS. 9–10)

What is measured, not presumed. The medium's population relaxation ($T_1 \approx 0.47 \mu\text{s}$ for the reference medium at 5.6 GHz) and coherence ($T_2^* \approx 0.1\text{--}0.3 \mu\text{s}$) are the known short times. A memory of hours-class retention on a continuously decaying observable at $L = 256$ levels requires $\tau_{\text{hold}} \geq 2L \times 3600 \text{ s} \approx 1.8 \times 10^6 \text{ s}$ (three weeks); at $L = 16$, $\approx 1.2 \times 10^5 \text{ s}$ (1.3 days). Those figures follow from the half-level rule and are stated so that every retention target in this disclosure is traceable to a required τ_{hold} .

Multi-timescale retention diagnostic (qualification). For each level k of the cell class, write, then record the spectroscopic observable (and, where fitted, the force-channel and temperature channels) over log-spaced windows from 10 ns to at least 10^4 s (extended toward 10^6 s for memory-class candidates). Fit the record as a sum of components with independently measured time constants: cavity ring-down τ_c (from loaded Q); population ring-down T_1 (from pulsed response and linewidth); thermal relaxation τ_{th} of body and mount (from temperature sensors and the absorbed-energy-matched detuned sham); and a residual component with free amplitude and time constant. Apply gates R1–R6 (Definition 14) to the residual. If all pass, enter $\tau_{\text{long}}(T, H_0, k)$ with uncertainty in the map and grade the class *memory-class*; if not, enter the upper bound on any residual amplitude and time constant, set $\tau_{\text{hold}} = T_1$, and grade the class *register-class*. Repeat at each operating temperature for species C.

Refresh-scheduled baseline (all grades). The refresh scheduler (718) sets $t_{\text{ref}} \leq s \cdot \tau_{\text{hold}} / (2L)$ from the map. A register-class cell is a multi-level resonant latch refreshed on the population timescale (for the reference medium, $L = 16$, $s = 0.5$: $t_{\text{ref}} \approx 7 \text{ ns}$, i.e., held under near-continuous re-drive at read power) and is a product of this disclosure for buffer, latch, and analog-register uses; a memory-class cell is refreshed at $s \cdot \tau_{\text{long}} / (2L)$. Both use the same architecture, map, and operating method; the grade changes only the scheduler parameter and the level count the map supports.

Cryogenic grade (species C; prospective). In embodiments in which τ_{hold} or τ_{long} is found at qualification to increase with decreasing temperature, operation at or below 77 K lengthens the refresh interval and may increase the supported level count through narrowed linewidth; the Arrhenius frame of

the recording art ($\tau = \tau_0 e^{\{KV/kT\}}$, $\tau_0 \approx 1$ ns: 40 kT ≈ 7 yr, 60 kT $\approx 3.6 \times 10^9$ yr, 0.69 kT per doubling) is cited only as the comparison frame for orientation stores. No barrier between winding levels is asserted; if the identification protocol finds a temperature dependence of τ_{long} consistent with an activated process, the fitted barrier is entered in the map as a measured quantity.

Endurance (design expectation and test). The write mechanism requires no structural change and no charge trapping in the medium; the expected wear mechanisms are thermal cycling of the mount and magnetostrictive fatigue at the holder. Endurance is qualified by cycling a cell through $\geq 10^9$ full-scale write/erase cycles within the thermal envelope while monitoring linewidth, level-0 reference, and full-scale observable; the class is rated at the cycle count at which any of these drifts beyond the map tolerance. "Substantially wear-free" in this disclosure means that rating exceeds the stated product life at the stated duty.

QUALIFICATION, CALIBRATION, AND DEPLOYMENT (FIG. 13, Tiers 2–3)

Factory qualification versus deployed operation. Qualification is performed once per cell class (and per addressed device) with the full instrumentation: the reference cavity or feed, a calibrated bias and gradient map, incident/reflected/transmitted power detectors, body and mount thermometry, an absorbed-energy-matched detuned sham, matched non-alignable and magnetic reference bodies, and, for the force-channel species, an independently supported force sensor with reaction-frame witness. Deployed operation uses the stored map, the runtime witnesses, and the reduced self-tests of the deployed operating method.

Empirical-map procedure. (a) *Acquire*: execute a randomized sequence of write levels, read powers, pulse parameters, bias points, and control states across the intended envelope; record all synchronized channels through the multi-timescale windows. (b) *Fit*: establish $\delta\phi$ and n_{turn} (from the observable's response to counted pulses), the readout map (observable $\rightarrow$ level, with monotonicity and uncertainty), the read-disturb threshold, τ_{hold} and the R1–R6 outcome, and, for the channel-partition species, the classification map. (c) *Validate*: held-out levels; replication (R6); artifact classification. (d) *Store*: write the retention and readout map, grade, refresh parameters, level count L , validity envelope, provenance, and safe-off parameters to the calibration memory. (e) *Ship and operate*. (f) *Update*: runtime zero and reference reads are logged; drift beyond tolerance shrinks the envelope, reduces L , shortens t_{ref} , or triggers recalibration.

Production-class qualification. A class map is established from a plurality of qualified cells of common design; an additional cell is accepted under the class map after an acceptance test (level-0 reference, full-scale observable, τ_{hold} spot check) agrees within tolerance.

What qualification does not presume. Level count, retention time, grade, readout resolution, response class, sign, and scaling with body size or count are outputs of measurement; each is absent from the map until measured.

Layered scope (FIG. 13). L0 raw synchronous observable (always reportable) → L1 mapped level index with uncertainty (inside the validity envelope) → L2 operation from the map. The acceptance gates and readout calibration stand between L0 and L1; the runtime validity gate between L1 and L2.

FREQUENCY-ADDRESSED WINDING-REGISTER MEMORY DEVICE (800) — FIGS. 7–8

Architecture. A common drive feed (870: coaxial or stripline transmission line) carries write, read, and comb tones from a locked source (860) to N cells ($872_1 \dots 872_N$) of species S, each coupled to the feed by a weakly coupled tap (874) and each biased to a distinct resonance address f_i by a **stepped, piecewise-uniform bias** (876): the conductor-free species uses a permanent-magnet yoke with per-cell shim elements or stepped pole pieces; the wired alternative uses per-cell trim coils carrying only static bias current. In either species there are **no per-cell signal conductors**—no word or bit lines through or to the storage medium; tones select cells by frequency. A device controller (880) holds the winding registers of all cells and the device's address map (875) and per-cell retention and readout maps.

Guard-band rule and property-box consistency. Address pitch Δf satisfies $\Delta f \geq 3\gamma\Delta H + B_p$, where ΔH is the occupied linewidth of the cell class and $B_p \approx 1/\tau_p$ is the write-pulse spectral width. Species S ($\Delta H \leq 0.5 \text{ Oe} = 1.4 \text{ MHz}$) with $\tau_p = 2 \mu\text{s}$ ($B_p = 0.5 \text{ MHz}$) requires $\Delta f \geq 4.7 \text{ MHz}$, satisfied by the 5 MHz reference pitch (1.79 Oe in bias). Species L at 10 Oe (28 MHz) is *not* array-consistent at that pitch and is not a species of this device except at a pitch $\geq 84 \text{ MHz}$.

Crosstalk architecture (FIG. 8). (i) *Stepped bias, not a linear gradient*: a linear gradient of 1.79 Oe per cell at 15 mm spacing is 1.19 Oe/cm and would broaden a 10 mm body by $\approx 1.2 \text{ Oe}$ (two thirds of the pitch); the stepped bias holds the field variation across each body to less than $\text{pitch}/(3\gamma)$. (ii) *Pulse bandwidth*: write pulses of width $\geq 1 \mu\text{s}$ ($\leq 1 \text{ MHz}$) at a 5 MHz pitch; 100 ns pulses (10 MHz) are excluded. (iii) *Dipolar budget and in-situ address calibration*: a 10 mm species-L body ($0.074 \text{ A}\cdot\text{m}^2$) produces $\approx 18 \text{ G}$ axially / $\approx 9 \text{ G}$ equatorially at 20 mm—several pitches—whereas a 1 mm species-S body ($7.4 \times 10^{-5} \text{ A}\cdot\text{m}^2$) produces $\approx 1.2 \text{ G}$ at 5 mm and $\approx 0.15 \text{ G}$ at 10 mm; the reference spacing is ≥ 10 body diameters, and the address map is calibrated in situ with all neighbours at level 0 and re-verified at full scale, the neighbour-write perturbation ($\Delta M_z/M_s \times \text{static dipolar shift}$) being recorded as a crosstalk term. (iv) *Guard cells*: unaddressed cells at the ends of the address band and, in dense devices, between address groups. (v) *Shielding partitions*: conductive or high-permeability partitions between cell groups on the reaction frame. (vi) *Weak coupling*: the coupling rate of each tap is held below the line's loss rate so that no hybrid doublet splits an address. (vii) *Comb block operations*: a frequency-comb waveform with one tone per addressed cell writes or refreshes a block; tone amplitudes are set per cell from the maps.

Device operating method. The device controller runs the deployed operating method per cell with frequency selection, schedules refresh per block by the most restrictive cell in the block, verifies by swept read of the whole band, and enters safe-off for the device when any cell's witness leaves the envelope.

SINK-GATED RESONANT POWER-SWITCHING DEVICE (900) AND RECEIVER-KEYED SIGNALING DEVICE (950) — FIGS. 11–12

Evidentiary status and qualification boundary. The POAMS-native sink-gating and receiver-keyed signaling species disclosed here are prophetic and unverified. Ordinary resonant coupling, passive-load modulation, and two-path apparatus dependence are compatibility examples only; none independently demonstrates source-side suppression before the conventional source-to-sink delay. The phrases "no signal emitted," "no transferable content," and "no separate enable signal" state the intended qualified operating condition, not an achieved result. A device is reported as POAMS-native only after the finite trial protocol below passes; otherwise it is reported as an extant-compatible resonant or load-modulated implementation.

Power-switching device (900). A source readiness apparatus (905) controlled by a source drive (903); a linkage gate (908) recording whether the source and sink readiness conditions co-sign a completed exchange; and a resonant sink (910) that, when presented and matched, completes the linkage and draws power. Delivery η in an optional coupled-resonator implementation at coupling coefficient k between resonators of quality factors Q_s and Q_k is $\eta = k^2 Q_s Q_k / (1 + \sqrt{(1 + k^2 Q_s Q_k)})^2$ (coupled-mode result); with $k = 0.05$ and $Q_s = Q_k = 10^3$, $k^2 Q_s Q_k = 2500$ and $\eta \approx 0.96$, so 10 W delivered requires ≈ 10.4 W at the source. **Standby.** In the POAMS-native embodiment standby delivery and exchange loss are zero because no hand-off is booked before completion. Optional source-resonator readiness hardware may dissipate locally; such apparatus loss is separately metered and is not energy in transit or delivery to the sink. **Transition.** The exchange is booked at completion; the instrument's finite registration or settling time is reported separately and is not treated as a propagating switching transient. **Exclusivity.** Only a load completing the linkage's resonance and coupling condition draws power; where additional address authentication is required, the source frequency-hops among stored addresses or requires a coded readiness sequence (below) before raising delivery above pilot.

Signaling device (950). On an established linkage the sink encodes symbols by modulating its own readiness (presented 962 / withdrawn 964 / detuned) and the source reads them from its local completion and hand-off ledger (968), the channel thereby being keyed from the receiving end—the inversion of transmit-keyed communication (nearest extant cousin: load modulation in passive backscatter tags). Symbols may also be conveyed as modulation of hand-off rate (Definition 24) and, for a qualified cell class with two response classes, as channel-response class. **Emission and enable condition.** Absent a completed linkage no signal or transferable content is emitted in the POAMS-native account; there is no separate enable signal from either end. Optional field probes, antenna monitors, or source-resonator telemetry may be used to bound artifacts or implement an extant-compatible species, but their readings do not constitute the native linkage or its ontology.

Receiver-keyed signaling qualification. Before native status is entered, the controller runs randomized and blinded READY/BLOCKED symbol blocks with synchronized source and sink clocks; records calibrated source-side power, current, resonator state, alternative-sink census, antenna/field-probe readings, and decoded symbols; and partitions the source record into pre-delay and post-delay bins using

an independently measured conventional delay. The protocol includes matched detuning, dummy-load, source-only, sink-only, cable/fixture, and alternative-sink controls. It reports trial counts, confidence intervals, detection floor, false-positive and false-negative rates, and a finite upper bound when no pre-delay effect is resolved. Native receiver-keyed operation is qualified only when a preregistered source-side readiness response occurs in the pre-delay bin, follows the blinded sink symbols with the specified fidelity, disappears in the controls, and all local apparatus and alternative-sink ledgers close. Failure does not defeat the resonant power-switching or extant-compatible load-modulation embodiments.

PROPHETIC EXAMPLES (prospective; every number follows from the stated species, pulse parameters, and rules; no result is represented as achieved)

P1 — memory-class cell target (conditional on qualification). A species-C cell (1 mm YIG sphere at 77 K) is written to $L = 256$ levels at $n_L = 10^4$ pulses per level, $2\ \mu\text{s}$ pulses at $\leq 10\ \text{mW}$ peak ($\leq 20\ \text{nJ}$ per pulse), 100 kHz repetition (2 mW average absorbed; 0.1 s per level; 25.6 s full scale). Adjacent levels at $\geq 5\sigma$ require an observable dynamic range of 62 dB and $\sigma_f \leq \Delta f_{\text{FS}}/1280$; for a prospective full-scale shift $\Delta f_{\text{FS}} = 10\ \text{MHz}$ (species D) that is $\sigma_f \leq 7.8\ \text{kHz}$, met with $\Delta f_{\text{line}} = 1.4\ \text{MHz}$ at $\text{SNR} \cdot \sqrt{M} \geq 180$ (e.g., SNR 30 dB and $M \geq 33$ sweeps). Hours-class retention is the design target *if* the identification protocol enters $\tau_{\text{long}} \geq 1.8 \times 10^6\ \text{s}$ at 77 K in the map; otherwise the cell ships register-class or at the level count and refresh interval the measured τ_{hold} supports. **P2 — force-channel read through containment (species L).** A 16-level 10 mm cell sealed in a copper enclosure with a filtered coaxial feedthrough on a 0.01 mg-force (98 nN) balance; read gradient 0.01 T/m applied during toggle windows; a retained $\Delta M_z/M_s$ of 1 % at full scale would give $\approx 7.4\ \mu\text{N}$, $\approx 0.46\ \mu\text{N}$ per level (≈ 4.7 balance floors); thermal artifact bounded at $\leq 5\ \text{nN}$ by the seven controls. The stored level is recovered by weighing alone, with no readout signal emitted by the cell; whether any level-dependent ΔF exists is the map's output. **P3 — 64-cell addressed device (species S).** Addresses 5.600–5.915 GHz at 5 MHz pitch (1.79 Oe steps by permanent-magnet shims), 1 mm spheres at 10 mm spacing (static neighbour shift $\approx 0.15\ \text{G} \approx 8\%$ of pitch, calibrated in situ), $2\ \mu\text{s}$ pulses, comb block write of 64 tones, swept verify of the 320 MHz band. **P4 — sink-gated switch.** 10 W delivered at $\eta \approx 0.96$ ($k = 0.05$, $Q = 10^3$ each); pilot standby dissipation $\leq 1\ \text{mW}$; transition $\approx Q_L/f$; standby radiated power below the listening-band target. **P5 — register-class latch.** A species-S cell whose qualification finds no long-retention channel: $L = 16$, $\tau_{\text{hold}} = T_1 \approx 0.47\ \mu\text{s}$, $t_{\text{ref}} \approx 7\ \text{ns}$ —a multi-level resonant latch under near-continuous refresh, offered as a product of the same architecture. **P6 — two-dimensional symbol (conditional).** A species-L cell whose classification map shows two drive-selectable response classes stores (winding depth, class): 4 bits + 1 bit per cell.

CLAIMS

1. A winding-register memory cell comprising: a body of ordered medium having an internal angular-momentum population alignable by a static field and having sufficient phase coherence; an alignment-field source configured to maintain a substantially uniform static alignment field across the body during write and hold; a coupling structure and a pulsed drive source configured to apply

counted resonant pulses at a located precession resonance of the aligned population, each pulse advancing the collective precession phase by a winding increment; a probe configured to read a winding observable of the body; a calibration memory storing a retention and readout map for the cell or for a qualified cell class to which the cell belongs, the map having been established by measurement and containing a relation between the winding observable and a level index with uncertainty, a read-disturb threshold, a measured hold time constant, a refresh interval derived from the hold time constant and the level count, a validity envelope, a provenance record, and safe-off parameters; and a controller holding a winding register and configured to write a selected one of at least three levels by applying a number of counted pulses fixed by the map, to read the level index from the winding observable through the map at a probe power below the read-disturb threshold, to refresh by reading and rewriting the returned level at the refresh interval, and to transition the cell to a safe-off state defined by the safe-off parameters when a validity condition of the map fails.

2. The cell of claim 1, wherein the winding observable is a center shift and a saturation depth of an absorption line of the body relative to a level-0 reference stored in the map.
3. The cell of claim 1, further comprising a force sensor supporting the body through a body load path separate from a reaction path carrying the coupling structure and the alignment-field source, and a read-gradient source configured to apply a calibrated force-axis field gradient to the body only during readout, wherein the winding observable is a differential support force of the body between drive-on and drive-off windows of a read-power drive toggle with the calibrated gradient applied, the cell emitting no readout signal.
4. The cell of claim 3, enclosed in electromagnetically opaque containment having a filtered drive feedthrough or an internal drive source, the level index being read through the containment by the force sensor.
5. The cell of claim 3, wherein the alignment-field source has a commanded sense, the controller reverses the sense in separate settled blocks, and the calibration memory stores a classification map assigning a response class from sense-odd and sense-even components of the differential support force, a second symbol dimension being written by selection of pulse width, bias point, or drive polarization and read as the response class when the map records at least two distinguishable drive-selectable classes for the cell class.
6. The cell of claim 1, wherein the level count supported by the map is at least sixteen.
7. The cell of claim 1, further comprising a cryogenic enclosure holding the body at or below 77 K, the retention and readout map containing the hold time constant and level count measured at that temperature.
8. The cell of claim 1, wherein the body is a polished single-crystal yttrium-iron-garnet sphere of diameter 0.2–2 mm having a resonance linewidth of at most 0.5 Oe at the operating frequency and 295 K and a resistivity of at least $10^{10} \Omega\text{-cm}$, or a sphere of diameter 2–15 mm having a linewidth of at most 10 Oe, or a disk or rod of aspect ratio at least 5:1 having a linewidth of at most 2 Oe.

9. The cell of claim 1, wherein the map records the level count as a function of the hold time constant such that the refresh interval does not exceed one half of the hold time constant divided by the level count, and records the cell class as register-class when the hold time constant equals the population relaxation time or as memory-class when a long-retention channel has passed stored identification gates.
10. A method of operating a winding-register memory cell having a body of ordered medium under a substantially uniform alignment field, a coupling structure, a pulsed drive source, a probe, and a calibration memory holding a measured retention and readout map, the method comprising: loading the map and verifying its provenance; performing a level-0 reference read; writing a selected level by applying to the body a number of counted resonant pulses fixed by the map for that level and recording the level in a winding register; verifying by reading a winding observable at a probe power below a read-disturb threshold stored in the map and comparing the returned level index with the winding register; holding under the alignment field; refreshing by reading and rewriting the returned level at a refresh interval not exceeding a stored fraction of the measured hold time constant divided by twice the level count; reading the level index with uncertainty from the map on demand; and transitioning to safe-off when a runtime witness leaves the validity envelope of the map.
11. The method of claim 10, further comprising erasing by applying counter-sense resonant drive of the committed count or a relaxation pulse comprising a bias-field excursion or a temperature pulse, and confirming erasure by a level-0 reference read.
12. The method of claim 10, wherein reading comprises applying a calibrated force-axis field gradient to the body only during drive-toggle windows, measuring the differential support force of the body, executing at least an off-resonance toggle control and a sense-reversal control, and mapping the differential force to the level index.
13. The method of claim 10, further comprising reducing the level count, shortening the refresh interval, or flagging recalibration when interleaved zero and reference reads drift beyond a tolerance stored in the map.
14. A frequency-addressed winding-register memory device comprising: a plurality of winding-register memory cells according to claim 1 coupled to a common drive feed by weakly coupled taps; a stepped, piecewise-uniform bias source setting each cell to a distinct resonance address, the field variation across each body being less than one third of the address pitch expressed in field; the address pitch being at least three times the occupied linewidth of the cells plus the spectral width of the write pulse; no per-cell signal conductor passing to the bodies; and a device controller holding the winding registers and an address map calibrated in situ and configured to select cells by tone frequency for write, read, and refresh.
15. The device of claim 14, wherein the bias source comprises a permanent-magnet yoke with per-cell shim elements or stepped pole pieces and no per-cell conductor of any kind.
16. The device of claim 14, further comprising guard cells at the ends of the address band and shielding partitions between groups of cells, the address map recording a neighbour-write crosstalk

term for each cell.

17. The device of claim 14, wherein the device controller writes or refreshes a block of cells by a frequency-comb waveform having one tone per addressed cell with amplitude set from that cell's map.
18. A method of operating a frequency-addressed winding-register memory device having a plurality of cells on a common feed at distinct bias-set resonance addresses, the method comprising: calibrating an address map in situ with all cells at level 0 and re-verifying it at full scale; writing a selected cell by tones at its address with pulse widths whose spectral width is less than the guard band; verifying by a swept read of the address band; refreshing each block at the shortest refresh interval among its cells; and transitioning the device to safe-off when any cell's runtime witness leaves its validity envelope.
19. A sink-gated resonant power-switching device comprising: a source readiness apparatus; a source drive configured to make a source-side exchange condition available; a linkage detector recording completion or noncompletion of a two-ended linkage; and a controller configured to enable delivery of resonant power to a resonant sink only upon detection of a completed linkage by presentation of a matched sink resonance, to disable delivery upon withdrawal or detuning of the sink resonance, to bound standby dissipation of optional readiness hardware separately from delivered power, and to report instrument registration time separately from the completion event, without mechanical contact.
20. The device of claim 19, wherein the controller raises delivery above the pilot level only after a stored coded readiness sequence is read from the linkage detector, or hops the source frequency among stored addresses.
21. A method of switching resonant power comprising: establishing a source readiness condition without booking delivery; detecting completion of a resonant linkage by presentation of a matched sink from a local completion-ledger observable of the source; enabling delivery of resonant power upon detection; and disabling delivery upon withdrawal or detuning of the sink resonance, no transferable content or separate enable signal being emitted before completion, without mechanical contact or arcing.
22. A receiver-keyed resonant signaling device comprising: a source readiness apparatus; a sink resonator having a readiness modulator configured to present, withdraw, or detune the sink resonance according to a symbol sequence; a completion-ledger detector at the source reading whether an exchange is completed and its booked hand-off rate; and a decoder configured to recover the symbol sequence from that local record, the channel being keyed from the receiving end, no separate enable signal being transmitted from either end, and no signal being emitted absent completion.
23. The device of claim 22, further comprising an optional antenna monitor, field probe, or source-resonator monitor used only to bound implementation artifacts while the sink is withdrawn, a reading of the monitor not being treated as transferred signal content.

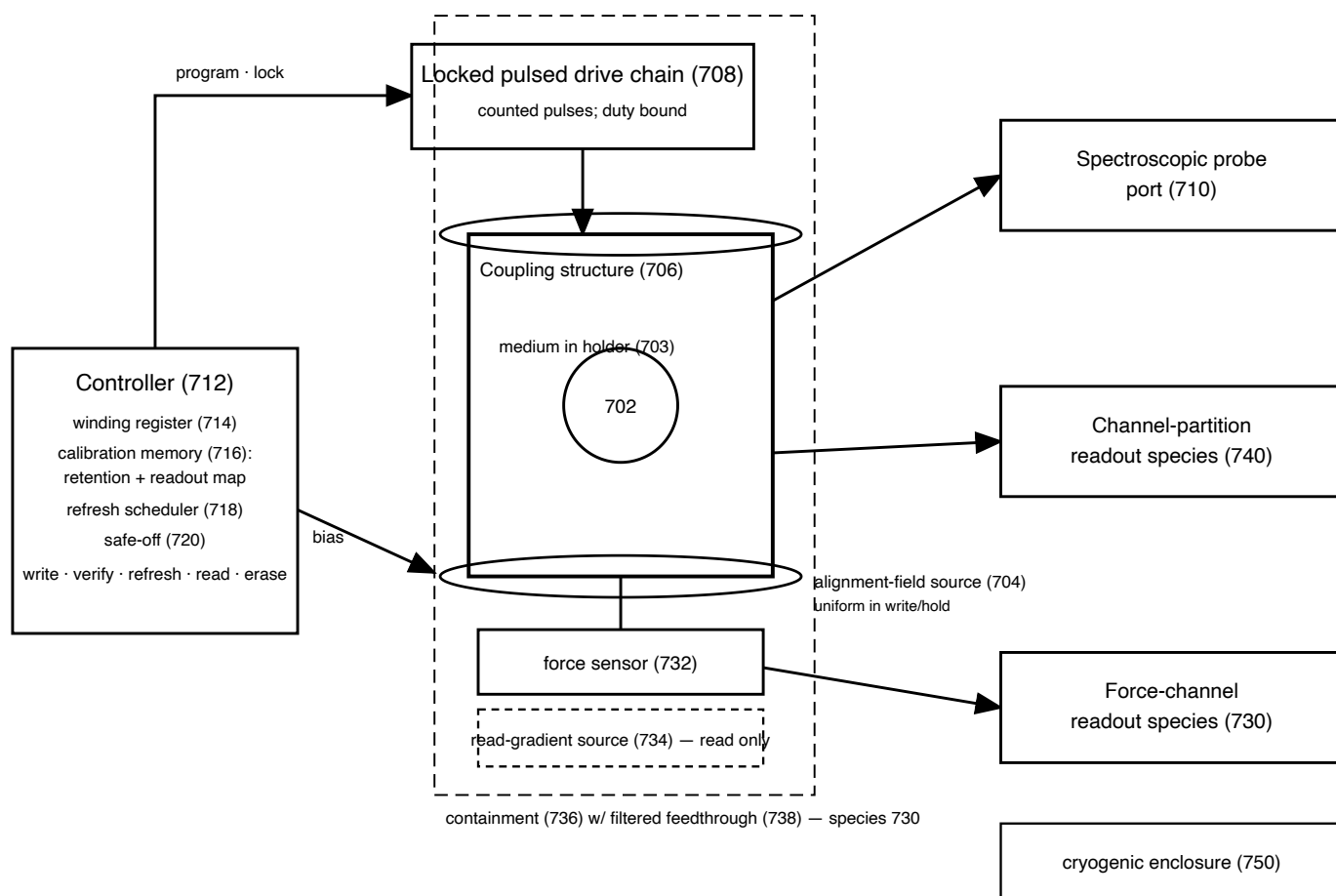
24. A signaling method on a resonant linkage between a source readiness apparatus and a sink resonator, comprising: establishing source readiness without booking delivery; encoding symbols at the sink by modulating the sink's readiness by presentation, withdrawal, or detuning of its resonance; reading the symbols at the source from its local completion and hand-off ledger; and transitioning between symbols when the sink changes readiness, whereby the channel is keyed from the receiving end and no signal content is emitted absent completion.
25. The method of claim 24, wherein symbols are further conveyed by modulation of hand-off rate, being the delivered power on the linkage expressed as a count rate of action quanta, or, for a source comprising a qualified winding-register cell class having at least two drive-selectable response classes, by channel-response class.
26. A method of qualifying a winding-register memory cell class comprising: writing each of a plurality of levels into a cell of the class by counted resonant pulses; recording a winding observable over logarithmically spaced windows from at most 10 ns to at least 10^4 s after each write; fitting the record as a sum of a cavity ring-down component, a population ring-down component, a thermal relaxation component, each with independently measured time constant, and a residual component; applying identification gates comprising time separation of the residual from the population and thermal components, resonance contrast against an absorbed-energy-matched detuned write, material contrast against a matched non-alignable body, monotonicity of the residual with committed count over at least three levels, erase contrast, and replication; entering in a retention and readout map a long-retention time constant with uncertainty when all gates pass and otherwise an upper bound and the population relaxation time as the hold time constant; establishing the relation between the winding observable and the level index with uncertainty and the read-disturb threshold; computing the refresh interval from the hold time constant and the level count; and recording the cell class as memory-class or register-class.
27. The method of claim 26, further comprising repeating the recording and fitting at a plurality of temperatures including at or below 77 K and entering the hold time constant and level count as functions of temperature.
28. The method of claim 26, further comprising, for a cell having a force sensor and a read-gradient source, recording the differential support force under a read-power drive toggle at each level and at each alignment sense, executing off-resonance, matched-body, sense-reversal, gradient-sweep, blinded level-interleave, toggle-frequency, and body-removed controls, and entering a force readout map and a classification map of response classes in the retention and readout map.

ABSTRACT

Winding-register memory cells, frequency-addressed memory devices, and sink-gated power-switching and receiver-keyed signaling devices on a resonantly driven ordered medium, with methods of operating and qualifying them. A cell holds a single-crystal body under a uniform alignment field in a tuned coupling structure; counted resonant pulses deposit a committed winding depth—whole turns of collective precession phase—read back non-destructively through a calibrated winding observable

(absorption-line shift and saturation depth; differential support force under a drive toggle with a calibrated read gradient, including through sealed containment; or channel-partition response class). The controller runs write, verify, hold, refresh, read, and erase from a measured retention and readout map whose hold time, level count, and register- or memory-class grade are qualification outputs; long retention is a hypothesis with disclosed identification gates. Arrays are frequency-addressed on one feed under a guard-band rule; linkage delivery is gated and keyed by the receiving resonator within bounded standby dissipation.

Winding-register memory cell (700) — medium, bias, locked pulsed drive, map, three readout species

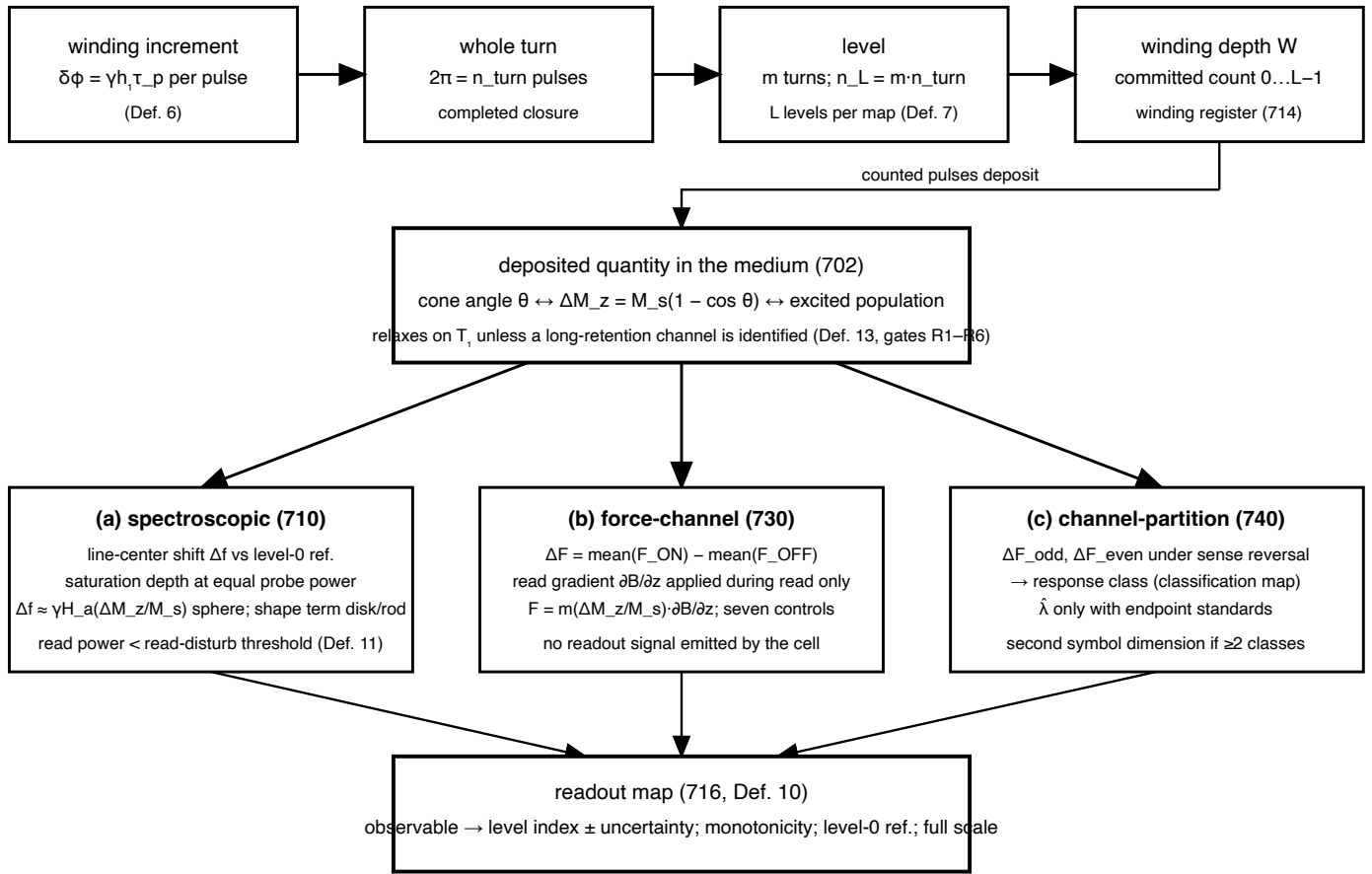


runtime witnesses: body/mount T, bias, gradient witness, absorbed power, lock, vibration

FIG. 1

Winding-register memory cell (700) — ordered medium in a coupling structure under a substantially uniform alignment field, locked pulsed drive, controller with winding register and calibration memory holding the retention and readout map, refresh scheduler and safe-off; spectroscopic, force-channel (with read-gradient source and containment feedthrough), and channel-partition readout species; cryogenic enclosure.

State variable and readout — increment → turn → level → committed register; three winding observables



W is a material state only to the extent the readout map returns it — that extent is measured at qualification, not presumed (Def. 8)

FIG. 2

State variable and readout definition — per-pulse winding increment, whole turn, level, and committed winding register; the deposited quantity in the medium; the three winding observables; and the readout map that returns the level index with uncertainty.

Write timing — counted pulses advance the committed winding depth one level per n_L pulses

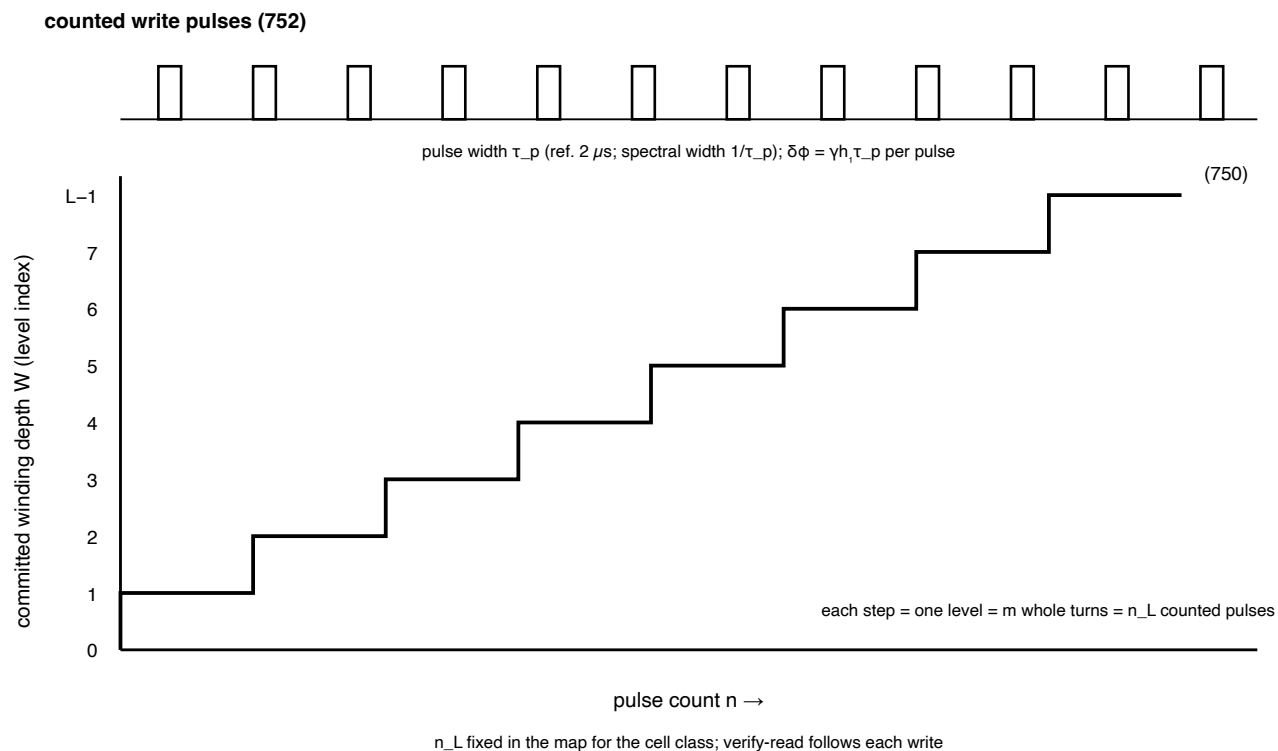
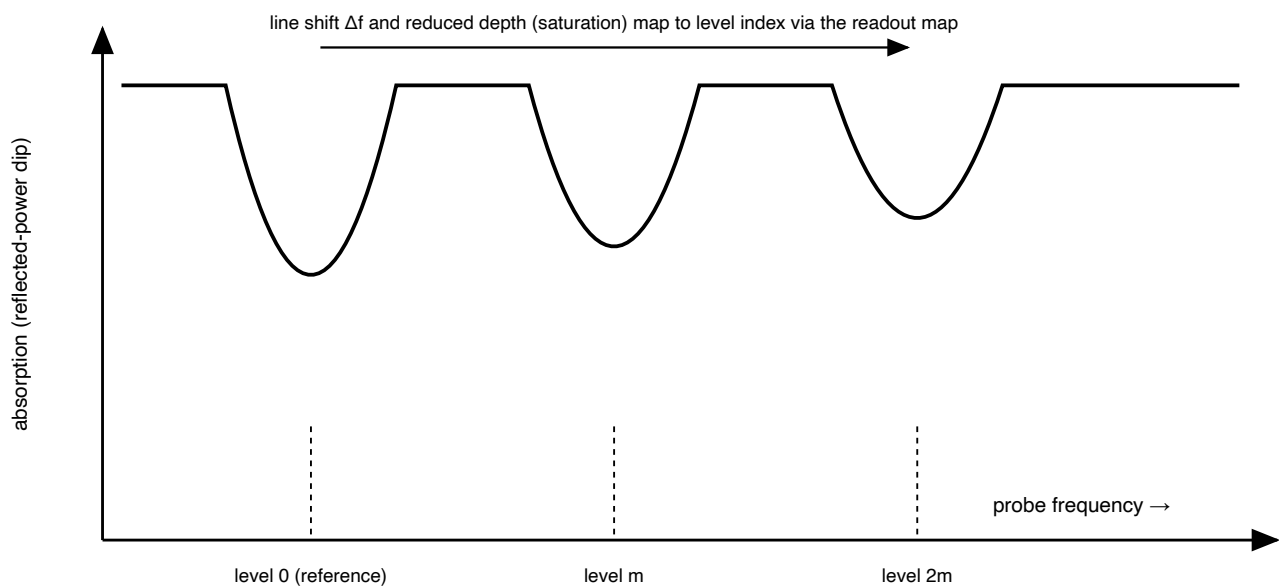


FIG. 3

Write timing — the committed winding depth W advances one level per $n_L = m \cdot n_{\text{turn}}$ counted pulses; level ladder $0 \dots L-1$ with L set by the retention and readout map.

Spectroscopic readout — line center and saturation depth versus stored level (755)



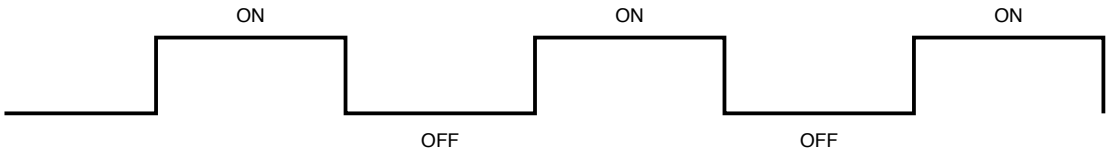
probe power below the read-disturb threshold (N_r reads change level by $< \frac{1}{2}$); $\sigma_f \leq \Delta f_{FS}/(5L)$ for adjacent levels at $\geq 5\sigma$
 sphere: $\Delta f \approx \gamma H_a (\Delta M_z / M_s) \approx 1.7 \text{ MHz per } 1\% M_s$ ($H_a \approx 60 \text{ Oe}$); disk/rod species D: shape term, one to two orders larger

FIG. 4

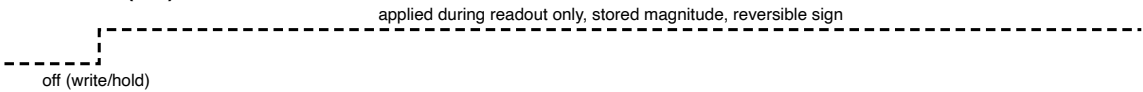
Spectroscopic readout — non-destructive probe of line-center shift and saturation depth at levels 0, m, 2m, mapped to the level index by the readout map at read power below the read-disturb threshold.

Force-channel readout — differential support force under read-power toggle with calibrated read gradient (760)

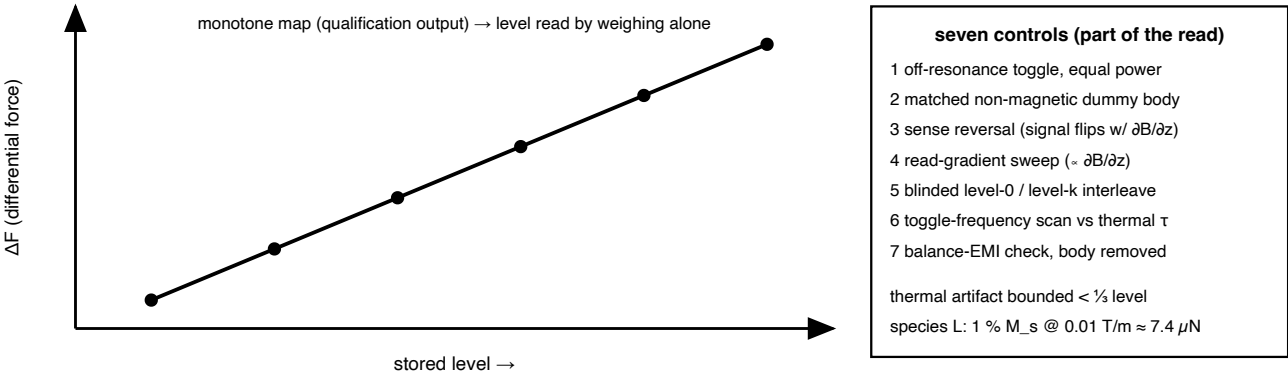
read-power drive toggle



read gradient $\partial B/\partial z$ (734)



$$\Delta F = \text{mean(ON)} - \text{mean(OFF)}, \text{ declared at } \geq 3\sigma; F = m_{\text{body}} \cdot (\Delta M_z / M_s) \cdot \partial B / \partial z$$



the cell emits no readout signal; read through EM-opaque containment (736) via filtered drive feedthrough (738)

FIG. 5

Force-channel readout — read-power drive toggle with the calibrated read gradient applied only during readout; ΔF mapped monotonically to the stored level by the readout map; the seven artifact controls that form part of the read.

Channel-partition symbol plane — stored symbol = (winding depth, response class) (765)

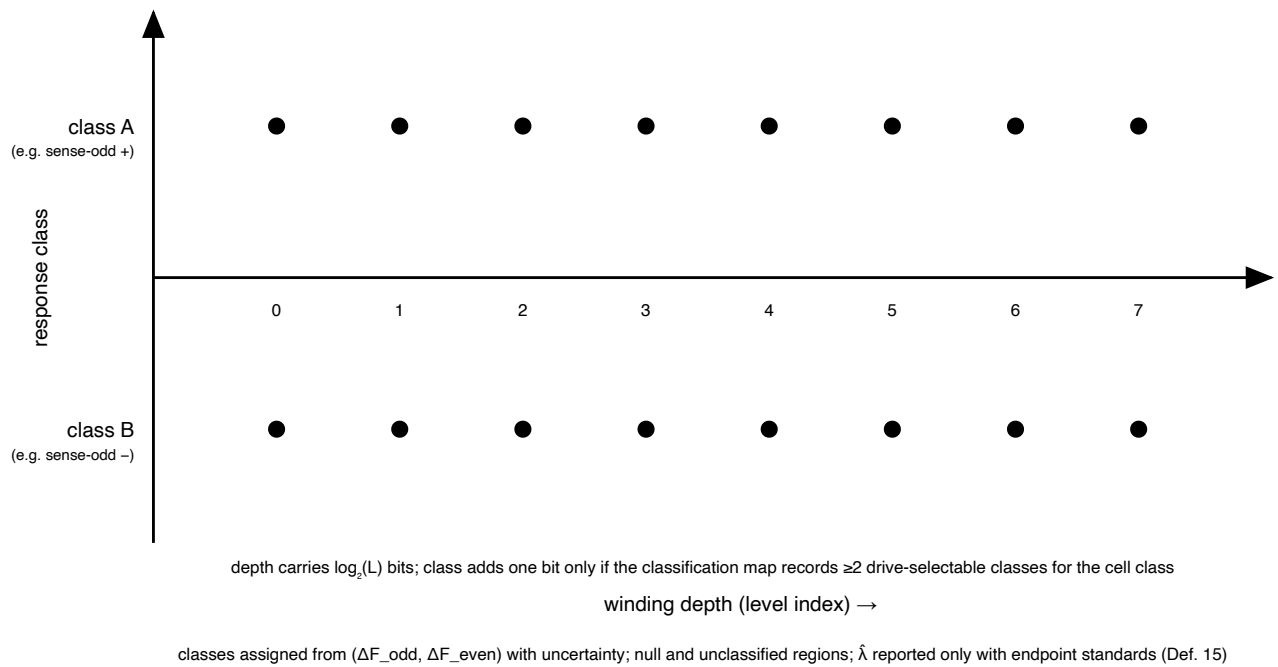
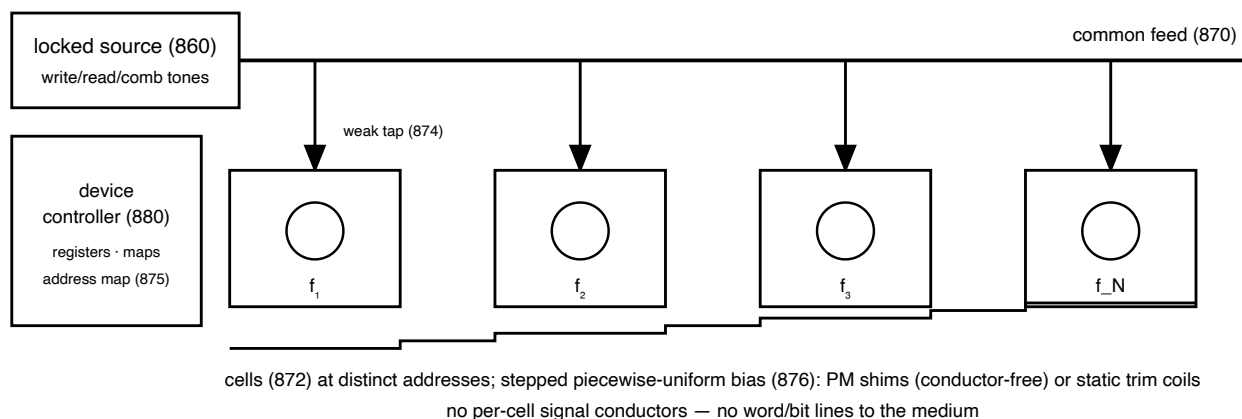


FIG. 6

Channel-partition symbol plane — each point is a storable symbol formed by winding depth and operational response class; the second dimension exists for a cell class only when qualification records two distinguishable drive-selectable classes.

Frequency-addressed winding-register memory device (800) — one common feed, stepped bias sets the address



address map (875) — in-situ calibrated

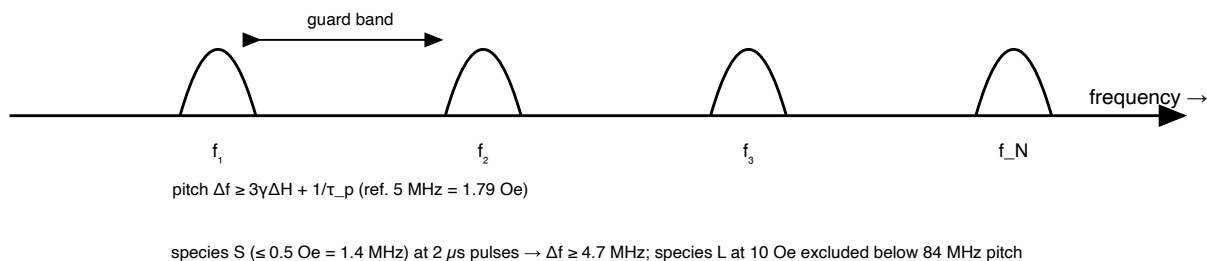


FIG. 7

Frequency-addressed winding-register memory device (800) — N cells on a shared feed via weakly coupled taps, each addressed by its stepped-bias resonance; device controller with registers, maps, and in-situ calibrated address map; guard-band rule.

Crosstalk architecture — guard-band rule, stepped bias, guard cells, partitions, dipolar budget, in-situ calibration

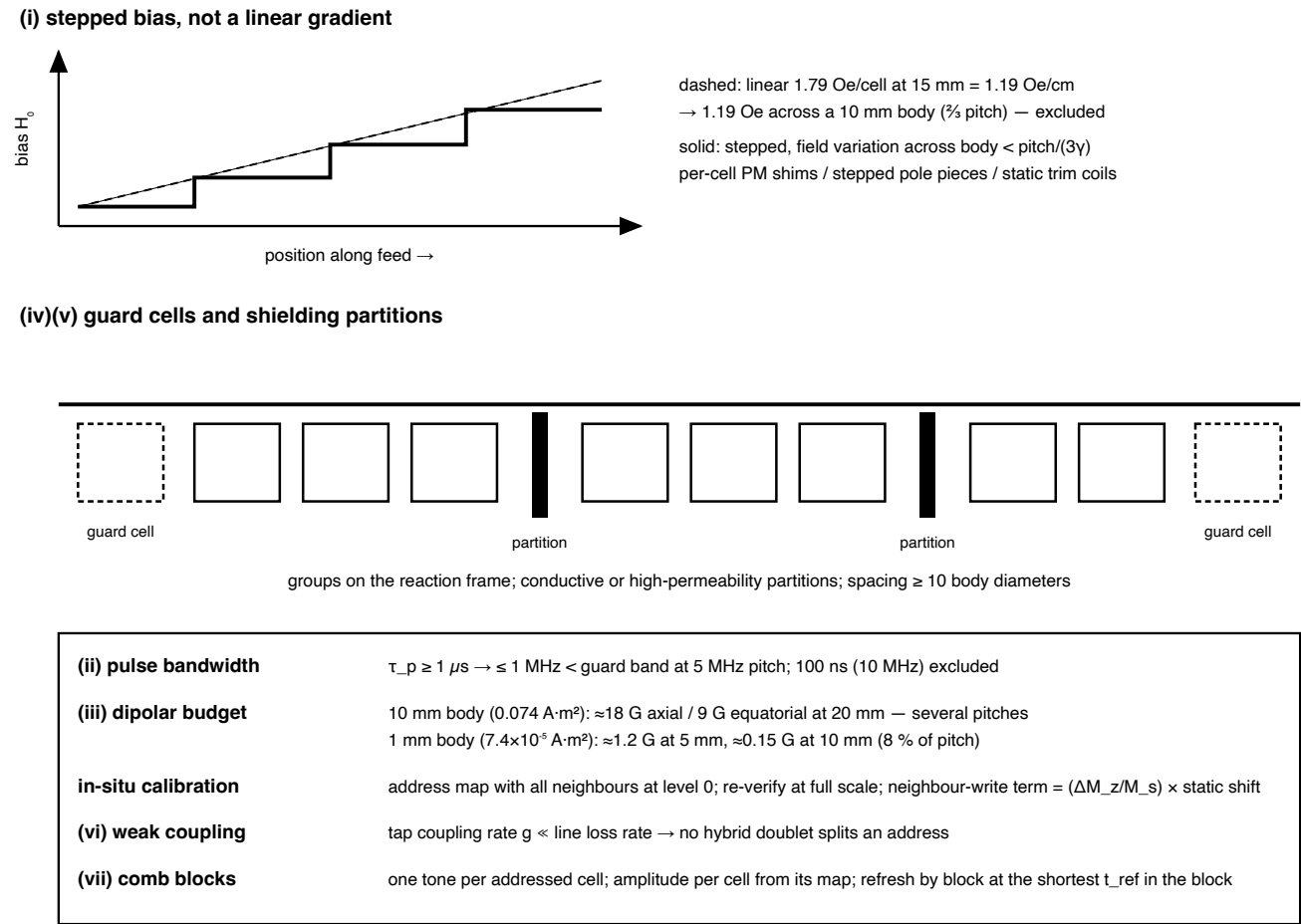


FIG. 8

Crosstalk architecture of the frequency-addressed device — stepped piecewise-uniform bias versus the excluded linear gradient; guard cells and shielding partitions; pulse-bandwidth, dipolar, weak-coupling, and comb-block rules; in-situ address calibration.

Retention (prospective) — retained level vs log time; register-class and memory-class grades (780)

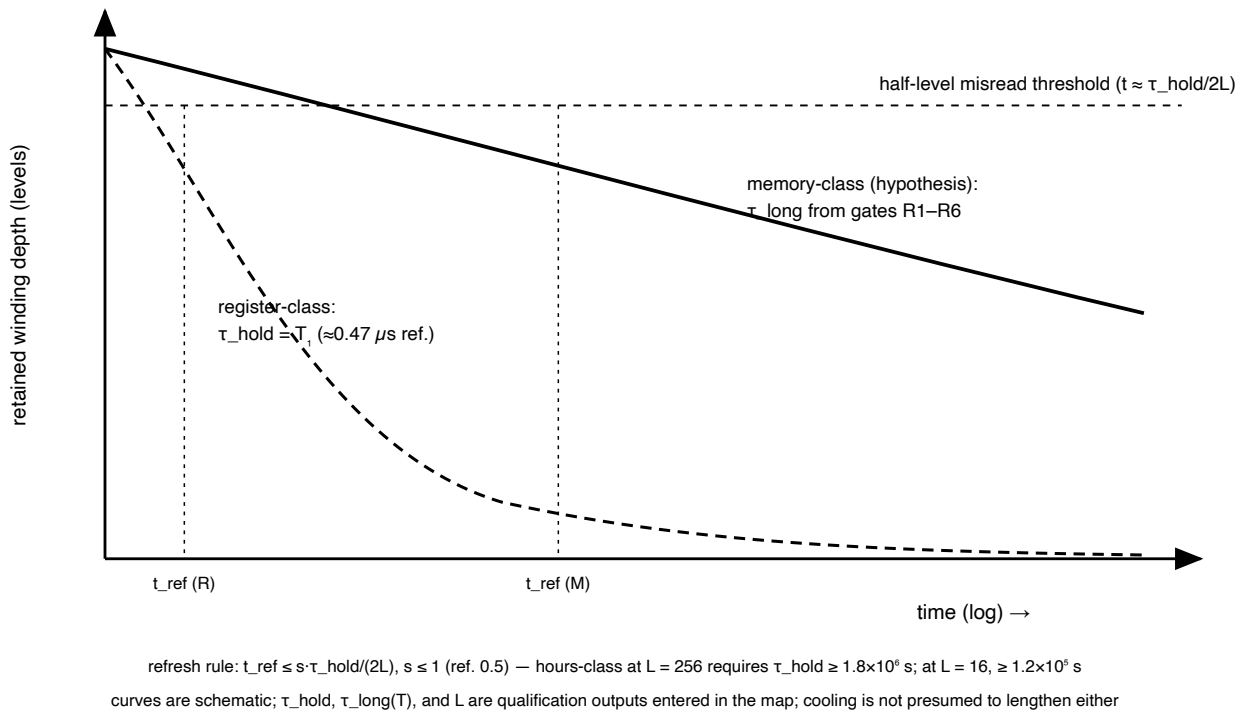


FIG. 9

Retention (prospective) — schematic decay of the retained level for a register-class cell (hold on the population relaxation) and a memory-class cell (identified long-retention channel); half-level misread threshold and the refresh intervals it sets.

Refresh-scheduled register and retention-identification protocol (qualification → map → scheduler)

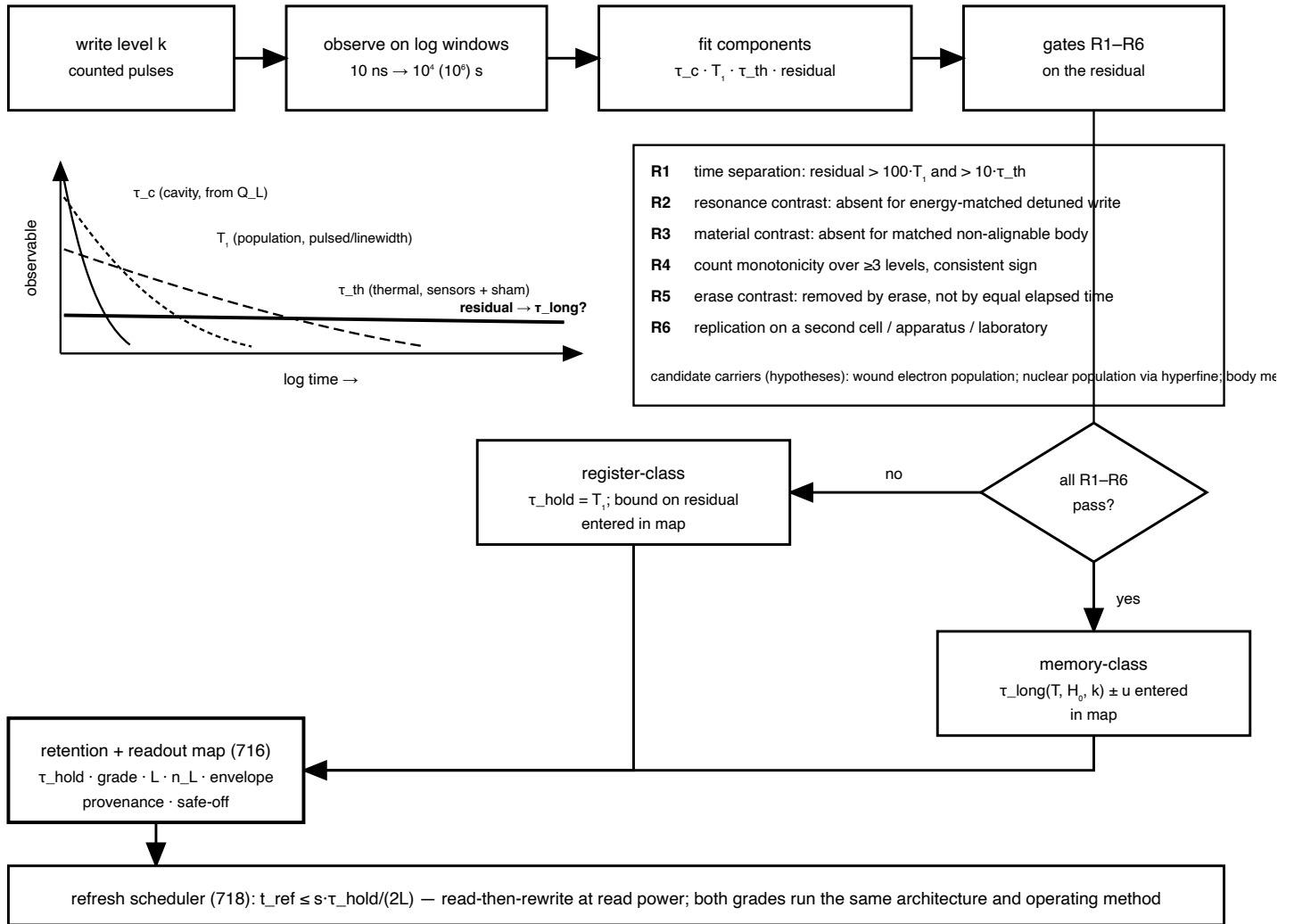
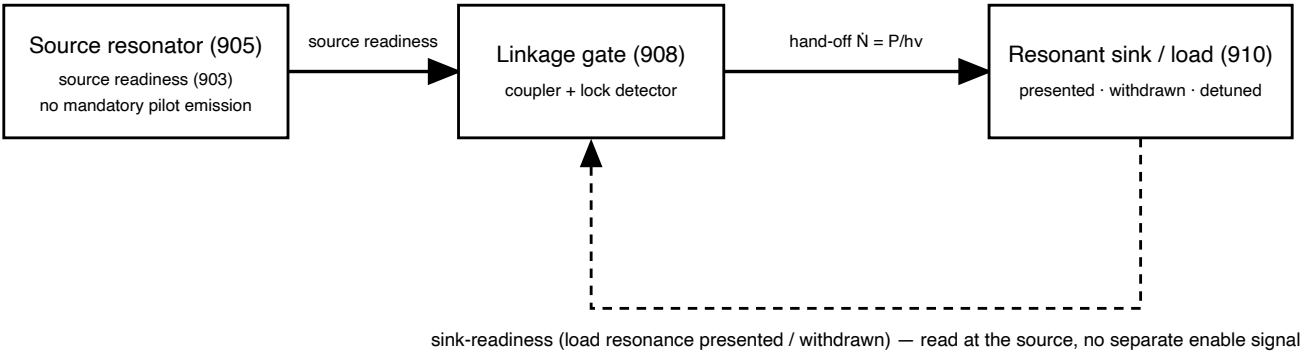


FIG. 10

Refresh-scheduled register and retention-identification protocol — multi-timescale observation after each write, component fit, gates R1–R6 on the residual, memory-class or register-class map entry, and the refresh scheduler that runs either grade from the map.

Sink-gated resonant power-switching device (900) — delivery only upon a completed resonant linkage



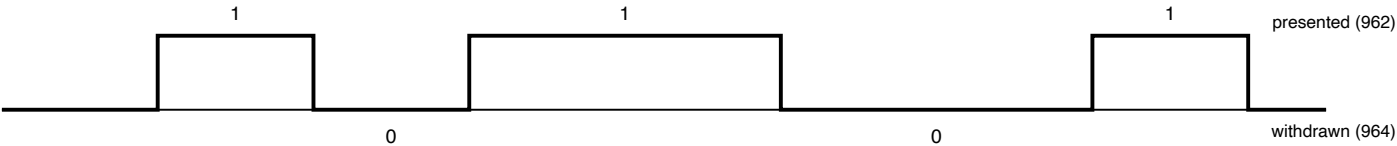
no completed linkage → no delivery; standby need not radiate transferable content; no contact, no arc
completion timing is recorded from the endpoint ledger, not presumed to be a propagation/reflection interval

delivered quanta and source debit are booked only on completed source–sink linkage

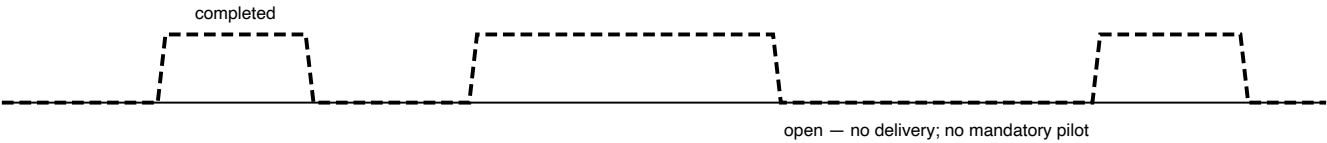
FIG. 11

Sink-gated resonant power-switching device (900) — source readiness, linkage gate, and resonant sink; delivery only upon completed linkage, without requiring a pilot emission or an in-transit energy store.

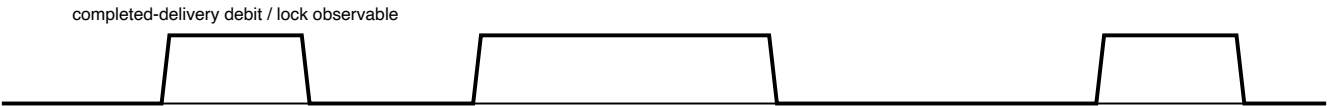
sink readiness (the receiver keys) — symbol stream (960), device 950



linkage state (966)



source delivery telemetry (968) — the source reads the symbols

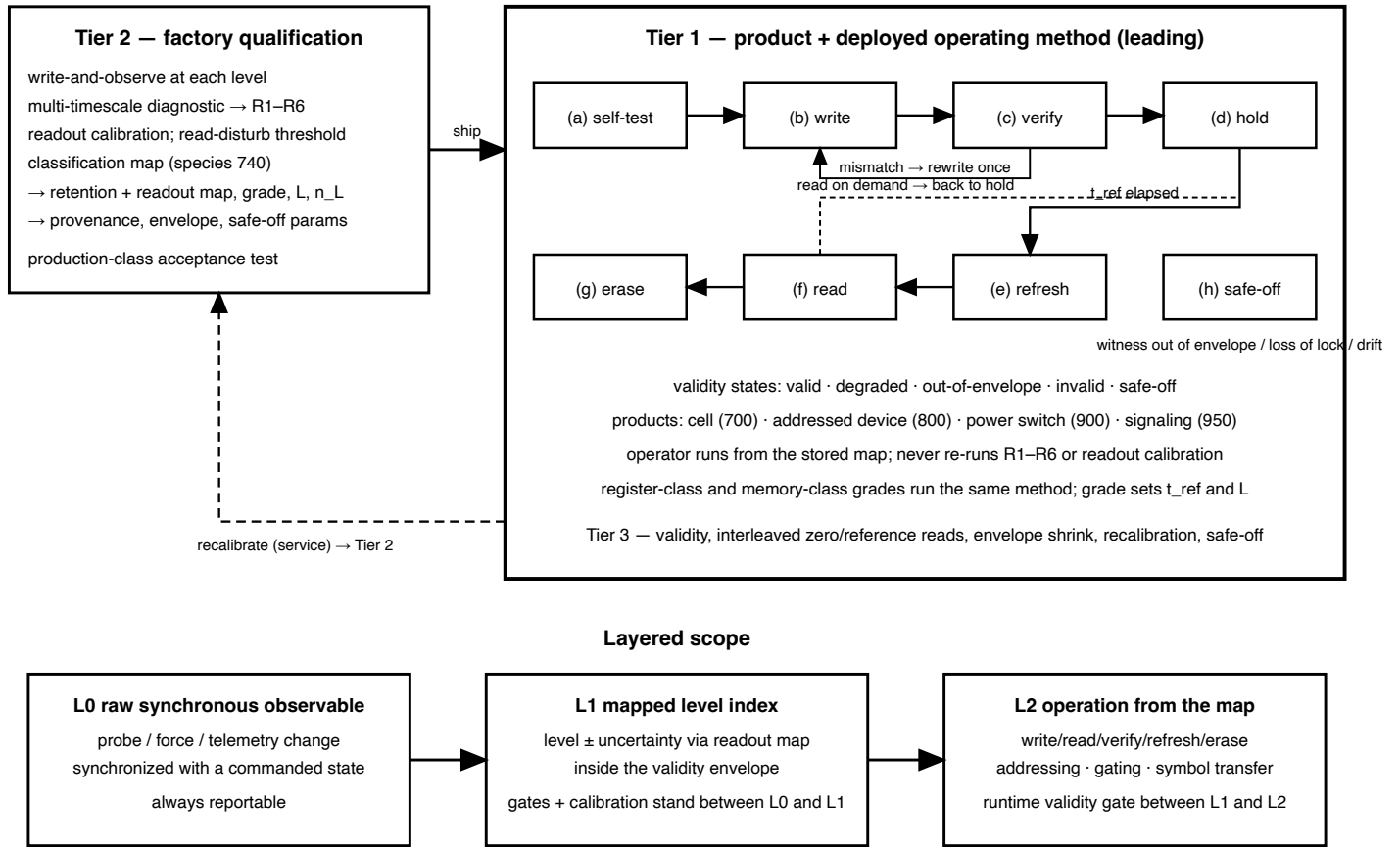


no separate enable signal from either end; the receiver keys each completed source–sink exchange (claims 22–25)
standby: no completed linkage and no emitted symbol-bearing content required
prophetic native species — endpoint timing and pre-delay response require blinded qualification

FIG. 12

Prophetic receiver-keyed resonant signaling device (950) — the sink encodes symbols by modulating readiness; native status requires randomized, blinded endpoint-timing qualification and closed artifact ledgers.

Three-tier architecture, deployed operating state machine, and layered scope



no layer presumes retention time, level count, readout resolution, response class, sign, or scaling beyond the validated map; the map admits a register-class grade

FIG. 13

Three-tier architecture and layered scope — Tier 1 products and the deployed operating state machine (a)–(h); Tier 2 factory qualification producing the retention and readout map and cell-class grade; Tier 3 validity, recalibration, and safe-off; layers L0 → L1 → L2.